

MAT101R01 – ENGINEERING MATHEMATICS – I

(Common for all branches of Engineering) Semester I

Unit III: Theory of Equations & Algebra

Relations between roots and coefficients – Complex and irrational roots – Roots in A.P., G.P., H.P. – Reciprocal equations – Transformation of equations – Symmetric functions – Multiple roots – Descartes' rule of signs – Stability analysis (Routh–Hurwitz) – Binomial, exponential and logarithmic series.

Lecture Notes by **Dr. S. Balaji****Contents**

1 Theory of equations: basic results	1
1.1 Relation between roots and coefficients	1
2 Finding the roots of an equation when some of its roots are known	2
3 Equations whose roots are in A.P., G.P. and H.P.	3
4 Reciprocal equations (R.E.)	5
4.1 Classification of R.E.	5
4.2 Problems	6
5 Transformation (formation) of equations	8
6 Symmetric functions of roots	13
6.1 Standard results on symmetric functions of the roots of a cubic	13
6.2 Problems	13
7 Additional problems in the solution of algebraic equations	14
7.1 Multiple roots	16
8 Descartes' rule of signs	17
9 Stability analysis of control systems (Routh–Hurwitz criterion)	17
9.1 Procedure involved in the Routh array method	18
9.2 Special cases in the Routh array	19
10 Algebra: Binomial series	21
10.1 Problems	22
10.2 Summation of series using binomial series	23
11 Exponential series	25
11.1 Problems	25
12 Logarithmic series	28
12.1 Problems	28
Assignment – III	30

1 Theory of equations: basic results

Generally an n th degree polynomial equation can be represented as

$$f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_{n-2}x^2 + a_{n-1}x + a_n = 0,$$

where $a_0, a_1, \dots, a_n$ are constants.

Result 1.1 (Fundamental theorem of algebra (statement only))

The above equation of n th degree will contain n roots satisfying the equation. Suppose they are $\alpha_1, \alpha_2, \dots, \alpha_n$; then the equation can be written as

$$f(x) = a_0(x - \alpha_1)(x - \alpha_2) \cdots (x - \alpha_n),$$

where $(x - \alpha_1), (x - \alpha_2), \dots, (x - \alpha_n)$ are the factors of the equation $f(x) = 0$.

Result 1.2 (Complex and irrational roots)

- (i) Complex (or imaginary) roots always appear in conjugate pairs, i.e. if $\alpha + i\beta$ is one root of $f(x) = 0$ then $\alpha - i\beta$ is also a root of $f(x) = 0$.
- (ii) Irrational roots also occur in conjugate pairs, i.e. if $\alpha + \sqrt{\beta}$ is one root of $f(x) = 0$ then $\alpha - \sqrt{\beta}$ is also a root of $f(x) = 0$. Similarly, if $\sqrt{\alpha} + \sqrt{\beta}$ is a root of $f(x) = 0$ then $\sqrt{\alpha} - \sqrt{\beta}$, $-\sqrt{\alpha} + \sqrt{\beta}$, $-\sqrt{\alpha} - \sqrt{\beta}$ are also roots of $f(x) = 0$.

1.1 Relation between roots and coefficients

Suppose for an n th degree algebraic polynomial equation $f(x) = a_0x^n + a_1x^{n-1} + \cdots + a_{n-1}x + a_n = 0$ the roots are $\alpha_1, \alpha_2, \dots, \alpha_n$. Then we can write $f(x)$ as

$$\begin{aligned} f(x) &= a_0(x - \alpha_1)(x - \alpha_2) \cdots (x - \alpha_n) \\ &= a_0 \left[x^n - (\alpha_1 + \alpha_2 + \cdots + \alpha_n)x^{n-1} + (\alpha_1\alpha_2 + \alpha_1\alpha_3 + \cdots + \alpha_{n-1}\alpha_n)x^{n-2} - \cdots + (-1)^n \alpha_1\alpha_2\alpha_3 \cdots \alpha_n \right] \\ f(x) &= a_0 \left[x^n - S_1x^{n-1} + S_2x^{n-2} - S_3x^{n-3} + \cdots + (-1)^n S_n \right], \end{aligned}$$

where S_r = sum of the products of the roots taken r at a time. Comparing coefficients on both the LHS and RHS:

$$x^{n-1}: a_1 = -a_0S_1 \Rightarrow S_1 = -\frac{a_1}{a_0}$$

$$\Rightarrow \boxed{S_1 = \alpha_1 + \alpha_2 + \cdots + \alpha_n = -\frac{a_1}{a_0} = (-1)^1 \frac{a_1}{a_0}}$$

$$x^{n-2}: a_2 = a_0S_2 \Rightarrow S_2 = \frac{a_2}{a_0}$$

$$\Rightarrow \boxed{S_2 = \alpha_1\alpha_2 + \alpha_1\alpha_3 + \cdots + \alpha_{n-1}\alpha_n = (-1)^2 \frac{a_2}{a_0}}$$

$$\text{Similarly, } x^{n-3}: a_3 = -a_0S_3 \Rightarrow S_3 = -\frac{a_3}{a_0} = (-1)^3 \frac{a_3}{a_0}$$

$$\Rightarrow \boxed{S_3 = \alpha_1\alpha_2\alpha_3 + \alpha_1\alpha_3\alpha_4 + \cdots + \alpha_{n-2}\alpha_{n-1}\alpha_n = (-1)^3 \frac{a_3}{a_0}}$$

$$\text{Constant: } a_n = a_0(-1)^n S_n \Rightarrow S_n = \frac{a_n}{(-1)^n a_0} = \frac{(-1)^n a_n}{a_0} \quad \left(\because \frac{1}{(-1)^n} = (-1)^n \right)$$

$$\Rightarrow \boxed{S_n = \alpha_1\alpha_2 \cdots \alpha_n = \frac{(-1)^n a_n}{a_0}}$$

Result 1.3 (Relations between roots and coefficients)

For $a_0x^n + a_1x^{n-1} + \dots + a_n = 0$ with roots $\alpha_1, \dots, \alpha_n$:

$$S_r = (-1)^r \frac{a_r}{a_0}, \quad r = 1, 2, \dots, n,$$

where S_1 = sum of the roots, S_n = product of the roots, and $S_2, S_3, \dots$ stand for the sum of the products of the roots taken 2 at a time, 3 at a time, etc.

Examples

(1) Consider a quadratic equation $ax^2 + bx + c = 0$; let the roots be α, β . Then

$$S_1 = \alpha + \beta = \frac{(-1)^1 b}{a} = -\frac{b}{a} \Rightarrow \boxed{\alpha + \beta = -\frac{b}{a}}, \quad S_2 = \alpha\beta = \frac{(-1)^2 c}{a} = \frac{c}{a} \Rightarrow \boxed{\alpha\beta = \frac{c}{a}}$$

(2) Consider a cubic equation $ax^3 + bx^2 + cx + d = 0$; let the roots be α, β, γ . Then

$$S_1 = \alpha + \beta + \gamma = \frac{(-1)^1 b}{a} \Rightarrow \boxed{\alpha + \beta + \gamma = -\frac{b}{a}}; \quad S_2 = \alpha\beta + \beta\gamma + \gamma\alpha = \frac{(-1)^2 c}{a} \Rightarrow \boxed{\alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}};$$

$$S_3 = \alpha\beta\gamma = \frac{(-1)^3 d}{a} \Rightarrow \boxed{\alpha\beta\gamma = -\frac{d}{a}}$$

(3) Consider a 4th degree equation $ax^4 + bx^3 + cx^2 + dx + e = 0$; let its roots be $\alpha, \beta, \gamma, \delta$. Then

$$S_1 = \alpha + \beta + \gamma + \delta = \frac{(-1)^1 b}{a} \Rightarrow \alpha + \beta + \gamma + \delta = -\frac{b}{a}$$

$$S_2 = \alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = \frac{(-1)^2 c}{a} = \frac{c}{a}$$

$$S_3 = \alpha\beta\gamma + \alpha\gamma\delta + \alpha\delta\beta + \beta\gamma\delta = \frac{(-1)^3 d}{a} = -\frac{d}{a}$$

$$S_4 = \alpha\beta\gamma\delta = \frac{(-1)^4 e}{a} = \frac{e}{a}.$$

2 Finding the roots of an equation when some of its roots are known

When we know some of its roots, we can form factors, and dividing the given equation through the factors we get the remaining equation (factor); solving the remaining factor we get the remaining roots also.

Problem 2.1

Solve the equation $x^4 + 2x^3 - 16x^2 - 22x + 7 = 0$, given that $2 + \sqrt{3}$ is one of its roots.

Solution. Given $f(x) = x^4 + 2x^3 - 16x^2 - 22x + 7 = 0$ and $x = 2 + \sqrt{3}$ is one root of $f(x) = 0 \Rightarrow x = 2 - \sqrt{3}$ is also a root of $f(x) = 0$ (irrational roots occur in conjugate pairs).

$$\begin{aligned} &\Rightarrow (x - (2 + \sqrt{3}))(x - (2 - \sqrt{3})) \text{ is a factor of } f(x) = 0 \\ &\Rightarrow ((x - 2) - \sqrt{3})((x - 2) + \sqrt{3}) \text{ is a factor of } f(x) = 0 \\ &\Rightarrow (x - 2)^2 - (\sqrt{3})^2 \text{ is a factor of } f(x) = 0 \\ &\Rightarrow (x^2 - 4x + 4 - 3) \Rightarrow (x^2 - 4x + 1) \text{ is a factor of } f(x) = 0. \end{aligned}$$

Now divide the given equation by this factor; use classical division:

$$\begin{array}{r|l}
 x^2 - 4x + 1 & x^2 + 6x + 7 \\
 & x^4 + 2x^3 - 16x^2 - 22x + 7 \\
 & \underline{x^4 - 4x^3 + x^2} \\
 & 6x^3 - 17x^2 - 22x \\
 & \underline{6x^3 - 24x^2 + 6x} \\
 & 7x^2 - 28x + 7 \\
 & \underline{7x^2 - 28x + 7} \\
 & 0
 \end{array}$$

$\Rightarrow$ The remaining factor is $x^2 + 6x + 7 = 0$. Roots of the remaining factor:

$$x = \frac{-6 \pm \sqrt{36 - 28}}{2} = \frac{-6 \pm \sqrt{8}}{2} = -3 \pm \sqrt{2}.$$

$\therefore$ Roots of the given equation are $\boxed{2 \pm \sqrt{3}, -3 \pm \sqrt{2}}$. ■

Exercise. Solve the equation $x^4 - 8x^3 + 34x^2 - 72x + 45 = 0$, given that $2 + i\sqrt{11}$ is one root.

3 Equations whose roots are in A.P., G.P. and H.P.

Problem 3.1

Solve the equation $2x^3 - 3x^2 - 11x + 6 = 0$ if their roots are in A.P.

Solution. Given $f(x) = 2x^3 - 3x^2 - 11x + 6 = 0$ and the roots of $f(x) = 0$ are in A.P. $(a, a+d, a+2d, a+3d, \dots)$. Let the roots be $a-d, a, a+d$. Then

$$S_1 = (a-d) + a + (a+d) = \frac{(-1)^1(-3)}{2} = \frac{3}{2} \Rightarrow 3a = \frac{3}{2} \Rightarrow \boxed{a = \frac{1}{2}} \text{ is one root of } f(x) = 0.$$

Divide the given equation by this root, i.e. by synthetic division:

$$\begin{array}{r|rrrr}
 \frac{1}{2} & 2 & -3 & -11 & 6 \\
 & 0 & 1 & -1 & -6 \\
 \hline
 & 2 & -2 & -12 & 0
 \end{array}$$

Remaining factor is $2x^2 - 2x - 12 = 0$, i.e. $x^2 - x - 6 = 0 \Rightarrow (x-3)(x+2) = 0 \Rightarrow x = 3, -2$. $\therefore$ Roots of the given $f(x) = 0$ are $-2, \frac{1}{2}, 3$, and they are in A.P. ■

Problem 3.2

Solve the equation $x^3 + 19x^2 + 114x + 216 = 0$ if its roots are in G.P.

Solution. Given $f(x) = x^3 + 19x^2 + 114x + 216 = 0$ and its roots are in G.P. $(a, ar, ar^2, ar^3, \dots \text{ or } \dots, \frac{a}{r^2}, \frac{a}{r}, a, ar, \dots)$. Let the roots be $\frac{a}{r}, a, ar$. Then

$$S_3 = \frac{a}{r} \cdot a \cdot ar = \frac{(-1)^3 216}{1} \Rightarrow a^3 = -216 = (-6)^3 \Rightarrow \boxed{a = -6} \text{ is one root of the given equation.}$$

By synthetic division,

$$\begin{array}{r|rrrr}
 -6 & 1 & 19 & 114 & 216 \\
 & 0 & -6 & -78 & -216 \\
 \hline
 & 1 & 13 & 36 & 0
 \end{array}
 \Rightarrow x^2 + 13x + 36 = 0 \Rightarrow (x+4)(x+9) = 0.$$

$\therefore$ The remaining roots are -4 and -9 , and the roots are $-9, -6, -4$. ■

Problem 3.3

Solve the equation $6x^3 - 11x^2 + 6x - 1 = 0$, whose roots are in H.P.

Solution. Given $f(x) = 6x^3 - 11x^2 + 6x - 1 = 0$ (1), whose roots are in H.P. $\therefore f\left(\frac{1}{x}\right) = 0$ has the roots in A.P.:

$$\begin{aligned}\therefore g(x) &= f\left(\frac{1}{x}\right) = 6\left(\frac{1}{x}\right)^3 - 11\left(\frac{1}{x}\right)^2 + 6\left(\frac{1}{x}\right) - 1 = 0 \quad \text{has the roots in A.P.} \\ \Rightarrow g(x) &= \frac{6}{x^3} - \frac{11}{x^2} + \frac{6}{x} - 1 = 0 \Rightarrow 6 - 11x + 6x^2 - x^3 = 0,\end{aligned}$$

i.e. $g(x) = x^3 - 6x^2 + 11x - 6 = 0$ (2) has the roots in A.P. Let the roots of $g(x)$ be $a - d, a, a + d$. From (2),

$$S_1 = (a - d) + a + (a + d) = \frac{(-1)^1(-6)}{1} \Rightarrow 3a = 6 \Rightarrow \boxed{a = 2}$$

The remaining roots are obtained by synthetic division:

$$\begin{array}{r|rrrr} 2 & 1 & -6 & 11 & -6 \\ & & 2 & -8 & 6 \\ \hline & 1 & -4 & 3 & 0 \end{array} \Rightarrow x^2 - 4x + 3 = 0 \Rightarrow x = 1 \text{ and } 3.$$

$\Rightarrow$ Roots of $g(x) = 0$ are 1, 2 and 3 (in A.P.). $\therefore$ Roots of $f(x) = 0$ are $1, \frac{1}{2}$ and $\frac{1}{3}$ (in H.P.). ■

Problem 3.4

Solve the equation $x^4 - 12x^3 + 14x^2 + 132x - 135 = 0$, given that the roots are in A.P.

Solution. Given $f(x) = x^4 - 12x^3 + 14x^2 + 132x - 135 = 0$ has the roots in A.P. Let the roots be $a - 3d, a - d, a + d, a + 3d$. Then by

$$S_1 = (a - 3d) + (a - d) + (a + d) + (a + 3d) = \frac{(-1)^1(-12)}{1} \Rightarrow 4a = 12 \Rightarrow \boxed{a = 3}$$

By

$$S_4 = (a - 3d)(a - d)(a + d)(a + 3d) = \frac{(-1)^4(-135)}{1} \Rightarrow (a^2 - 9d^2)(a^2 - d^2) = -135 \Rightarrow (9 - 9d^2)(9 - d^2) = -135$$

$$\Rightarrow 81 - 9d^2 - 81d^2 + 9d^4 = -135 \Rightarrow 9d^4 - 90d^2 + 81 + 135 = 0 \Rightarrow 9d^4 - 90d^2 + 216 = 0.$$

$$\text{Let } d^2 = t: 9t^2 - 90t + 216 = 0 \Rightarrow t^2 - 10t + 24 = 0 \Rightarrow (t - 6)(t - 4) = 0 \Rightarrow t = 4, 6.$$

$$t = 4 \Rightarrow d^2 = 4 \Rightarrow \boxed{d = \pm 2}; \quad t = 6 \Rightarrow d^2 = 6 \Rightarrow d = \pm\sqrt{6}.$$

When $d = \pm\sqrt{6}$ we get $a - d = 3 \mp \sqrt{6}$, $a + d = 3 \pm \sqrt{6}$, but these values do not satisfy the given equation $f(x) = 0$ (they are also inconsistent with $S_2 = 14$). $\therefore$ We must have $d = \pm 2$ and $a = 3$. Then the roots $a - 3d, a - d, a + d, a + 3d$ are: $d = 2 \Rightarrow -3, 1, 5, 9$; $d = -2 \Rightarrow 9, 5, 1, -3$.

$\therefore$ Roots are $-3, 1, 5, 9$ (in A.P.). ■

Problem 3.5

Solve the equation $x^3 - 13x^2 + 15x + 189 = 0$, given that two of its roots differ by 2.

Solution. Given $f(x) = x^3 - 13x^2 + 15x + 189 = 0$ has two of its roots differing by 2. Let α, β, γ be its roots and let $\boxed{\alpha - \beta = 2} \Rightarrow \boxed{\alpha = 2 + \beta}$. Then by

$$S_1 = \alpha + \beta + \gamma = \frac{(-1)^1(-13)}{1} \Rightarrow (2 + \beta) + \beta + \gamma = 13 \Rightarrow \boxed{2\beta + \gamma = 11}$$

Now

$$S_2 = \alpha\beta + \beta\gamma + \gamma\alpha = \frac{(-1)^2 15}{1}$$

$$\begin{aligned}
&\Rightarrow (2 + \beta)\beta + \beta(11 - 2\beta) + (11 - 2\beta)(2 + \beta) = 15 \\
&\Rightarrow 2\beta + \beta^2 + 11\beta - 2\beta^2 + 22 + 11\beta - 4\beta - 2\beta^2 = 15 \\
&\Rightarrow -3\beta^2 + 20\beta + 22 - 15 = 0 \\
&\Rightarrow -3\beta^2 + 20\beta + 7 = 0 \quad \text{or} \quad 3\beta^2 - 20\beta - 7 = 0 \\
&\Rightarrow 3\beta^2 - 21\beta + \beta - 7 = 0 \Rightarrow 3\beta(\beta - 7) + (\beta - 7) = 0 \Rightarrow (3\beta + 1)(\beta - 7) = 0,
\end{aligned}$$

so $\beta = 7$ or $\beta = -\frac{1}{3}$. $\therefore \alpha = 2 + \beta$ and $\gamma = 11 - 2\beta$.

When $\beta = 7$: $\alpha = 9$ and $\gamma = -3$. When $\beta = -\frac{1}{3}$: $\alpha = \frac{5}{3}$ and $\gamma = \frac{35}{3}$. $\therefore$ Roots are either 9, 7, -3 or $\frac{5}{3}, -\frac{1}{3}, \frac{35}{3}$. Obviously $x = -3$ satisfies the equation $f(x) = 0$ (whereas $\frac{5}{3} \cdot (-\frac{1}{3}) \cdot \frac{35}{3} \neq -189$).

$\therefore$ Roots are $9, 7, -3$. ■

4 Reciprocal equations (R.E.)

Definition 4.1 (Reciprocal equation)

An equation $f(x) = 0$ is said to be a **reciprocal equation** if both α and $\frac{1}{\alpha}$ are the roots of the equation $f(x) = 0$, i.e. if $\alpha_1, \alpha_2, \dots, \alpha_n$ are the roots of a reciprocal equation $f(x) = 0$, then $\frac{1}{\alpha_1}, \frac{1}{\alpha_2}, \dots, \frac{1}{\alpha_n}$ are also roots of $f(x) = 0$.

E.g. (1) $2x^2 - 5x + 2 = 0$ has both 2 and $\frac{1}{2}$ as its roots; it is an R.E.

(2) $2x^3 + 3x^2 - 3x - 2 = 0$ has 1, -2 and $-\frac{1}{2}$ as its roots, hence it is an R.E.

Result 4.1 (Condition for a reciprocal equation)

An equation $f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_{n-2}x^2 + a_{n-1}x + a_n = 0$ is said to be an R.E. if and only if

$$a_{n-r} = \pm a_r, \quad 0 \leq r \leq n, \quad \text{i.e.} \quad a_n = \pm a_0; \quad a_{n-1} = \pm a_1; \quad a_{n-2} = \pm a_2; \quad \dots$$

Remark. In an R.E., if $a_r = a_{n-r}$ then it is called an R.E. of the **first type**. If $a_r = -a_{n-r}$ then it is called an R.E. of the **second type**.

4.1 Classification of R.E.

- (1) An R.E. of first type and even degree is called a **Standard Reciprocal Equation (SRE)**.
- (2) An R.E. of first type and odd degree has $x = -1$ as one of its roots, and if we divide the equation by the factor $(x + 1)$, the given R.E. becomes an SRE.
- (3) An R.E. of second type and odd degree has $x = 1$ as one of its roots, and if we divide by $(x - 1)$, the given R.E. becomes an SRE.
- (4) An R.E. of second type and even degree has both $x = \pm 1$ as its roots, and if we divide by $(x + 1)(x - 1) = (x^2 - 1)$, the given R.E. becomes an SRE.

Generally an SRE can be solved by dividing the whole equation by the variable of half of the highest degree, i.e. if x^6 is the highest degree variable, then divide the whole equation by the variable x^3 .

4.2 Problems

Problem 4.1

Solve the equation $6x^4 - 25x^3 + 37x^2 - 25x + 6 = 0$.

Solution. Given $f(x) = 6x^4 - 25x^3 + 37x^2 - 25x + 6 = 0$ (1) is a first type and even degree R.E. Hence it is a Standard Reciprocal Equation (SRE). $\therefore$ Divide (1) by x^2 :

$$\begin{aligned}(1) &\Rightarrow 6x^2 - 25x + 37 - \frac{25}{x} + \frac{6}{x^2} = 0 \\ &\Rightarrow 6\left(x^2 + \frac{1}{x^2}\right) - 25\left(x + \frac{1}{x}\right) + 37 = 0.\end{aligned}\quad (2)$$

Let $\boxed{x + \frac{1}{x} = y} \Rightarrow \left(x + \frac{1}{x}\right)^2 = y^2 \Rightarrow x^2 + \frac{1}{x^2} + 2x \cdot \frac{1}{x} = y^2 \Rightarrow \boxed{x^2 + \frac{1}{x^2} = y^2 - 2}$. Substituting these in (2):

$$\begin{aligned}&\Rightarrow 6(y^2 - 2) - 25y + 37 = 0 \Rightarrow 6y^2 - 12 - 25y + 37 = 0 \\ &\Rightarrow 6y^2 - 25y + 25 = 0 \quad \text{is a quadratic equation} \\ &\Rightarrow 6y^2 - 15y - 10y + 25 = 0 \Rightarrow 3y(2y - 5) - 5(2y - 5) = 0 \\ &\Rightarrow (3y - 5)(2y - 5) = 0 \Rightarrow \boxed{y = \frac{5}{3}, \frac{5}{2}}\end{aligned}$$

Case (i): when $y = \frac{5}{3}$.

$$\begin{aligned}x + \frac{1}{x} = \frac{5}{3} &\Rightarrow \frac{x^2 + 1}{x} = \frac{5}{3} \Rightarrow 3x^2 + 3 = 5x \Rightarrow 3x^2 - 5x + 3 = 0 \\ &\Rightarrow x = \frac{5 \pm \sqrt{25 - 36}}{6} = \frac{5 \pm \sqrt{-11}}{6} = \boxed{\frac{5 \pm i\sqrt{11}}{6}}\end{aligned}$$

Case (ii): when $y = \frac{5}{2}$.

$$\begin{aligned}x + \frac{1}{x} = \frac{5}{2} &\Rightarrow \frac{x^2 + 1}{x} = \frac{5}{2} \Rightarrow 2x^2 - 5x + 2 = 0 \Rightarrow 2x^2 - 4x - x + 2 = 0 \\ &\Rightarrow 2x(x - 2) - (x - 2) = 0 \Rightarrow (2x - 1)(x - 2) = 0 \Rightarrow \boxed{x = 2, \frac{1}{2}}\end{aligned}$$

$\therefore$ Roots are $2, \frac{1}{2}, \frac{5 \pm i\sqrt{11}}{6}$. ■

Problem 4.2

Solve $6x^5 + 11x^4 - 33x^3 - 33x^2 + 11x + 6 = 0$.

Solution. Given $f(x) = 6x^5 + 11x^4 - 33x^3 - 33x^2 + 11x + 6 = 0$ (1) is a first type odd degree R.E. $\therefore x = -1$ is one of its roots of $f(x) = 0$. $\therefore$ Divide $f(x) = 0$ by the factor $x + 1$ (or use synthetic division):

$$\begin{array}{r|rrrrrr} -1 & 6 & 11 & -33 & -33 & 11 & 6 \\ & & -6 & -5 & 38 & -5 & -6 \\ \hline & 6 & 5 & -38 & 5 & 6 & 0 \end{array}$$

$\therefore$ The remaining factor $6x^4 + 5x^3 - 38x^2 + 5x + 6 = 0$ (2) is an SRE. $\therefore$ Divide (2) by x^2 :

$$\Rightarrow 6x^2 + 5x - 38 + \frac{5}{x} + \frac{6}{x^2} = 0 \Rightarrow 6\left(x^2 + \frac{1}{x^2}\right) + 5\left(x + \frac{1}{x}\right) - 38 = 0.$$

Let $\boxed{x + \frac{1}{x} = y} \Rightarrow \boxed{x^2 + \frac{1}{x^2} = y^2 - 2}$. $\therefore$ The above equation becomes

$$\begin{aligned}6(y^2 - 2) + 5y - 38 &= 0 \Rightarrow 6y^2 + 5y - 50 = 0 \Rightarrow 6y^2 - 15y + 20y - 50 = 0 \\ &\Rightarrow 3y(2y - 5) + 10(2y - 5) = 0 \Rightarrow (3y + 10)(2y - 5) = 0 \Rightarrow y = -\frac{10}{3}, \frac{5}{2}.\end{aligned}$$

Case (i): when $y = -\frac{10}{3}$.

$$x + \frac{1}{x} = -\frac{10}{3} \Rightarrow 3x^2 + 10x + 3 = 0 \Rightarrow 3x^2 + 9x + x + 3 = 0 \Rightarrow 3x(x + 3) + (x + 3) = 0$$

$$\Rightarrow (3x+1)(x+3) = 0 \Rightarrow \boxed{x = -3, -\frac{1}{3}}$$

Case (ii): when $y = \frac{5}{2}$.

$$x + \frac{1}{x} = \frac{5}{2} \Rightarrow 2x^2 - 5x + 2 = 0 \Rightarrow 2x^2 - 4x - x + 2 = 0 \Rightarrow 2x(x-2) - (x-2) = 0 \Rightarrow (2x-1)(x-2) = 0 \Rightarrow \boxed{x = 2, \frac{1}{2}}$$

$\therefore$ Roots are $x = -1, -3, -\frac{1}{3}, 2, \frac{1}{2}$. ■

Problem 4.3

Solve the equation $2x^5 - 15x^4 + 37x^3 - 37x^2 + 15x - 2 = 0$.

Solution. Given $f(x) = 2x^5 - 15x^4 + 37x^3 - 37x^2 + 15x - 2 = 0$ (1) is an R.E. of 2nd type and odd degree. $\therefore x = 1$ is one root of $f(x) = 0$. $\therefore$ Divide (1) by $x - 1$:

$$\begin{array}{r|rrrrrr} 1 & 2 & -15 & 37 & -37 & 15 & -2 \\ & 0 & 2 & -13 & 24 & -13 & 2 \\ \hline & 2 & -13 & 24 & -13 & 2 & 0 \end{array}$$

$\Rightarrow$ The remaining factor $2x^4 - 13x^3 + 24x^2 - 13x + 2 = 0$ (2) is an SRE. Dividing (2) by x^2 :

$$\Rightarrow 2x^2 - 13x + 24 - \frac{13}{x} + \frac{2}{x^2} = 0 \Rightarrow 2\left(x^2 + \frac{1}{x^2}\right) - 13\left(x + \frac{1}{x}\right) + 24 = 0.$$

Let $\boxed{x + \frac{1}{x} = y} \Rightarrow \boxed{x^2 + \frac{1}{x^2} = y^2 - 2}:$

$$\Rightarrow 2(y^2 - 2) - 13y + 24 = 0 \Rightarrow 2y^2 - 13y + 20 = 0 \Rightarrow 2y^2 - 8y - 5y + 20 = 0$$

$$\Rightarrow 2y(y - 4) - 5(y - 4) = 0 \Rightarrow (2y - 5)(y - 4) = 0 \Rightarrow \boxed{y = \frac{5}{2}, 4}$$

Case (i): when $y = \frac{5}{2}$. $x + \frac{1}{x} = \frac{5}{2} \Rightarrow 2x^2 - 5x + 2 = 0 \Rightarrow \boxed{x = 2, \frac{1}{2}}$.

Case (ii): when $y = 4$. $x + \frac{1}{x} = 4 \Rightarrow x^2 - 4x + 1 = 0 \Rightarrow x = \frac{4 \pm \sqrt{16-4}}{2} = 2 \pm \sqrt{3} \Rightarrow \boxed{x = 2 \pm \sqrt{3}}$.

$\therefore$ Roots are $x = 1, 2, \frac{1}{2}, 2 \pm \sqrt{3}$. ■

Problem 4.4

Solve $6x^6 - 5x^5 - 44x^4 + 44x^2 + 5x - 6 = 0$.

Solution. Given $f(x) = 6x^6 - 5x^5 - 44x^4 + 0x^3 + 44x^2 + 5x - 6 = 0$ is an R.E. of 2nd type, even degree. $\therefore x = \pm 1$ are roots of $f(x) = 0$. $\therefore$ Divide $f(x) = 0$ by $x \pm 1$:

$$\begin{array}{r|rrrrrrr} 1 & 6 & -5 & -44 & 0 & 44 & 5 & -6 \\ & 0 & 6 & 1 & -43 & -43 & 1 & 6 \\ \hline -1 & 6 & 1 & -43 & -43 & 1 & 6 & 0 \\ & 0 & -6 & 5 & 38 & 5 & -6 & \\ \hline & 6 & -5 & -38 & -5 & 6 & 0 & \end{array}$$

$\Rightarrow$ The remaining factor $6x^4 - 5x^3 - 38x^2 - 5x + 6 = 0$ (2) is an SRE. Dividing (2) by x^2 :

$$6x^2 - 5x - 38 - \frac{5}{x} + \frac{6}{x^2} = 0 \Rightarrow 6\left(x^2 + \frac{1}{x^2}\right) - 5\left(x + \frac{1}{x}\right) - 38 = 0.$$

Let $\boxed{x + \frac{1}{x} = y} \Rightarrow \boxed{x^2 + \frac{1}{x^2} = y^2 - 2}:$

$$\Rightarrow 6(y^2 - 2) - 5y - 38 = 0 \Rightarrow 6y^2 - 5y - 50 = 0 \Rightarrow 6y^2 + 15y - 20y - 50 = 0$$

$$\Rightarrow 3y(2y + 5) - 10(2y + 5) = 0 \Rightarrow (3y - 10)(2y + 5) = 0 \Rightarrow y = \frac{10}{3}, -\frac{5}{2}.$$

Case (i): when $y = \frac{10}{3}$. $x + \frac{1}{x} = \frac{10}{3} \Rightarrow 3x^2 - 10x + 3 = 0 \Rightarrow (3x - 1)(x - 3) = 0 \Rightarrow \boxed{x = 3, \frac{1}{3}}.$

Case (ii): when $y = -\frac{5}{2}$. $x + \frac{1}{x} = -\frac{5}{2} \Rightarrow 2x^2 + 5x + 2 = 0 \Rightarrow (2x + 1)(x + 2) = 0 \Rightarrow \boxed{x = -2, -\frac{1}{2}}.$

$\therefore$ Roots are $x = \pm 1, 3, \frac{1}{3}, -2, -\frac{1}{2}$. ■

5 Transformation (formation) of equations

In this section we form a new transformed equation $g(x) = 0$ from the equation $f(x) = 0$, where the roots of $g(x) = 0$ are functions of the roots of $f(x) = 0$, i.e. if $\alpha_1, \alpha_2, \dots, \alpha_n$ are the roots of $f(x) = 0$, we form a new equation $g(x) = 0$ whose roots are $\phi(\alpha_1, \alpha_2, \dots, \alpha_n)$.

Type I: Equation whose roots are k times the roots of $f(x) = 0$ (k a constant)

Result 5.1

The coefficients of the new equation are obtained by multiplying the coefficients $a_0, a_1, \dots, a_n$ of $f(x) = 0$ by $1, k, k^2, \dots, k^n$ respectively; this gives $g(x) = 0$.

Problem 5.1

Form the equation whose roots are 10 times the roots of the equation $x^3 + 3x^2 + x - 4 = 0$.

Solution. Given $f(x) = x^3 + 3x^2 + x - 4 = 0$. To get a new equation $g(x) = 0$ whose roots are 10 times the roots of $f(x) = 0$, multiply the coefficients of $f(x)$, i.e. 1, 3, 1 and -4 , by 1, 10, 10^2 and 10^3 . Now the coefficients of $g(x) = 0$ are 1, 30, 100, -4000

$$\Rightarrow \text{the new equation is } \boxed{g(x) = x^3 + 30x^2 + 100x - 4000 = 0},$$

whose roots are 10 times the roots of $f(x) = 0$. ■

Type II: Equation whose roots are the reciprocals of the roots of $f(x) = 0$

Result 5.2

The new equation is $g(x) = f\left(\frac{1}{x}\right) = 0$.

Problem 5.2

Form the equation whose roots are the reciprocals of the roots of $x^3 + x^2 - x - 1 = 0$.

Solution. Given $f(x) = x^3 + x^2 - x - 1 = 0$. Then the new equation $g(x) = 0$, whose roots are the reciprocal roots of $f(x) = 0$, is given by

$$g(x) = f\left(\frac{1}{x}\right) = \left(\frac{1}{x}\right)^3 + \left(\frac{1}{x}\right)^2 - \frac{1}{x} - 1 = 0 \Rightarrow 1 + x - x^2 - x^3 = 0 \Rightarrow \boxed{g(x) = x^3 + x^2 - x - 1 = 0}$$

(the given equation is itself a reciprocal equation). ■

Type III: Equation whose roots are the negatives of the roots of $f(x) = 0$ **Result 5.3**

The new equation is $g(x) = f(-x) = 0$.

Problem 5.3

Form the equation whose roots are the negatives of the roots of $x^3 + 3x^2 - 2x + 1 = 0$.

Solution. Given $f(x) = x^3 + 3x^2 - 2x + 1 = 0$. To get $g(x) = 0$ whose roots are the negative roots of $f(x) = 0$:

$$g(x) = f(-x) = (-x)^3 + 3(-x)^2 - 2(-x) + 1 = 0 \Rightarrow -x^3 + 3x^2 + 2x + 1 = 0 \Rightarrow \boxed{g(x) = x^3 - 3x^2 - 2x - 1 = 0}$$

■

Type IV: Diminishing (or increasing) the roots of $f(x) = 0$ **Result 5.4**

To form the new equation whose roots are the roots of $f(x) = 0$ diminished by h , divide $f(x) = 0$ successively by the factor $x - h$ (or use synthetic division successively by h). The successive remainders (last one first) give the coefficients of the new equation $g(x) = 0$, whose roots are $\alpha_1 - h, \alpha_2 - h, \dots, \alpha_n - h$, where $\alpha_1, \dots, \alpha_n$ are the roots of $f(x) = 0$.

To increase the roots of the equation by h , divide the whole equation successively by the factor $x + h$ (or by $-h$ using synthetic division).

Problem 5.4

Diminish the roots of the equation $x^3 + x^2 + x - 100 = 0$ by 4.

Solution. Divide $f(x) = x^3 + x^2 + x - 100 = 0$ (roots α, β, γ) successively by $h = 4$ using synthetic division:

4	1	1	1	-100
	0	4	20	84
4	1	5	21	-16
	0	4	36	
4	1	9	57	
	0	4		
	1	13		

Then the new equation is

$$\boxed{g(x) = x^3 + 13x^2 + 57x - 16 = 0},$$

whose roots are $\alpha - 4, \beta - 4, \gamma - 4$, obtained by diminishing the roots of $f(x) = 0$ by 4.

(By classical division: $x^3 + x^2 + x - 100 = (x - 4)(x^2 + 5x + 21) - 16$, then $x^2 + 5x + 21 = (x - 4)(x + 9) + 57$, and $x + 9 = (x - 4) \cdot 1 + 13$.) ■

Problem 5.5

Increase the roots of the equation $4x^5 - 2x^3 + 7x - 3 = 0$ by 2.

Solution. Divide (synthetically) by -2 the given equation $f(x) = 4x^5 + 0x^4 - 2x^3 + 0x^2 + 7x - 3 = 0$:

-2	4	0	-2	0	7	-3
	0	-8	16	-28	56	-126
-2	4	-8	14	-28	63	-129
	0	-8	32	-92	240	
-2	4	-16	46	-120	303	
	0	-8	48	-188		
-2	4	-24	94	-308		
	0	-8	64			
-2	4	-32	158			
	0	-8				
	4	-40				

∴ The transformed equation is

$$4x^5 - 40x^4 + 158x^3 - 308x^2 + 303x - 129 = 0,$$

and this equation has the roots which are increased by 2 the roots of $f(x) = 0$. ■

Formation of an equation without the second term

Result 5.5

Suppose we have the equation $a_0x^n + a_1x^{n-1} + \dots + a_n = 0$. To form an equation without the second term, we have to diminish the roots of the above equation by

$$h = -\frac{a_1}{na_0}.$$

Problem 5.6

Form an equation without the second term by diminishing the equation $4x^5 - 40x^4 + 158x^3 - 308x^2 + 303x - 129 = 0$ using the necessary h value.

Solution. Given $f(x) = 4x^5 - 40x^4 + 158x^3 - 308x^2 + 303x - 129 = 0$ ($n = 5$, $a_0 = 4$, $a_1 = -40$). To get an equation without the 2nd term, we have to diminish $f(x) = 0$ by

$$h = -\frac{a_1}{na_0} = \frac{-(-40)}{5(4)} = 2,$$

i.e. diminish by $h = 2$:

2	4	-40	158	-308	303	-129
	0	8	-64	188	-240	126
2	4	-32	94	-120	63	-3
	0	8	-48	92	-56	
2	4	-24	46	-28	7	
	0	8	-32	28		
2	4	-16	14	0		
	0	8	-16			
2	4	-8	-2			
	0	8				
	4	0				

The required equation is $4x^5 - 2x^3 + 7x - 3 = 0$ (which has no x^4 term; compare with the previous problem). ■

Type V: Formation of an equation whose roots are functions of the roots of $f(x) = 0$ **Result 5.6**

Under this, the new equation $g(y) = 0$ can be constructed by writing $y = \phi(\alpha)$ (using the relations S_1, S_2, S_3 wherever necessary). Then replace α by x and solve to get $x = h(y)$; substituting this in the equation $f(x) = 0$, we get the new equation $g(y) = 0$.

Problem 5.7

If α, β, γ are the roots of $x^3 + px^2 + qx + r = 0$, then form the equations whose roots are (i) $\alpha + \beta, \beta + \gamma, \gamma + \alpha$
(ii) $\alpha^2, \beta^2, \gamma^2$ (iii) $\frac{\alpha}{\beta + \gamma}, \frac{\beta}{\gamma + \alpha}, \frac{\gamma}{\alpha + \beta}$.

Solution. Given $f(x) = x^3 + px^2 + qx + r = 0$ has α, β, γ as its roots $\Rightarrow x = \alpha, x = \beta, x = \gamma$ satisfy $f(x) = 0$, and

$$S_1 = \alpha + \beta + \gamma = \frac{(-1)^1 p}{1} = -p, \quad S_2 = \alpha\beta + \beta\gamma + \gamma\alpha = \frac{(-1)^2 q}{1} = q, \quad S_3 = \alpha\beta\gamma = \frac{(-1)^3 r}{1} = -r.$$

(i) Roots $\alpha + \beta, \beta + \gamma, \gamma + \alpha$. We have to construct $g(y) = 0$ whose roots are $\alpha + \beta, \beta + \gamma, \gamma + \alpha$, i.e. $y = \alpha + \beta, y = \beta + \gamma, y = \gamma + \alpha$ satisfy $g(y) = 0$. Now consider $y = \alpha + \beta$ satisfies $g(y) = 0 \Rightarrow y = -p - \gamma$ ($\because$ from S_1) satisfies $g(y) = 0 \Rightarrow \gamma = -p - y$ satisfies $f(x) = 0$. $\therefore$ Substitute $x = -(p + y)$ in $f(x) = 0$:

$$\begin{aligned} &\Rightarrow (-(p + y))^3 + p(-(p + y))^2 + q(-(p + y)) + r = 0 \\ &\Rightarrow -(p^3 + y^3 + 3p^2y + 3py^2) + p(p^2 + y^2 + 2py) - qp - qy + r = 0 \\ &\Rightarrow -y^3 + y^2(-3p + p) + y(-3p^2 + 2p^2 - q) + (-p^3 + p^3 - qp + r) = 0 \\ &\Rightarrow -y^3 - 2py^2 - (p^2 + q)y + (r - qp) = 0 \\ &\Rightarrow \boxed{y^3 + 2py^2 + (p^2 + q)y + (qp - r) = 0} \end{aligned}$$

is the required equation whose roots are $\alpha + \beta, \beta + \gamma$ and $\gamma + \alpha$.

(ii) Roots $\alpha^2, \beta^2, \gamma^2$. To get $g(y) = 0$ whose roots are $\alpha^2, \beta^2, \gamma^2$: $y = \alpha^2, y = \beta^2, y = \gamma^2$ satisfy $g(y) = 0$. Now consider $y = \alpha^2$ satisfies $g(y) = 0 \Rightarrow \alpha = \sqrt{y}$ satisfies $f(x) = 0$. Substitute $x = \sqrt{y}$ in $f(x) = 0$. But

$$f(x) = x^3 + px^2 + qx + r = 0 \text{ i.e. } x^3 + qx = -(px^2 + r) \Rightarrow x(x^2 + q) = -(px^2 + r).$$

Squaring, $x^2(x^2 + q)^2 = (px^2 + r)^2$. Now substitute $x = \sqrt{y}$ in the above equation:

$$\begin{aligned} &\Rightarrow y(y + q)^2 = (py + r)^2 \Rightarrow y(y^2 + q^2 + 2qy) = p^2y^2 + r^2 + 2pyr \\ &\Rightarrow \boxed{y^3 + y^2(2q - p^2) + y(q^2 - 2pr) - r^2 = 0} \end{aligned}$$

is the required equation whose roots are $\alpha^2, \beta^2, \gamma^2$.

(iii) Roots $\frac{\alpha}{\beta + \gamma}, \frac{\beta}{\gamma + \alpha}, \frac{\gamma}{\alpha + \beta}$. Let $y = \frac{\alpha}{\beta + \gamma}$ satisfy the new equation $g(y) = 0$, i.e. $y = \frac{\alpha}{-p - \alpha}$ (from S_1) satisfies $g(y) = 0$

$$\Rightarrow y(p + \alpha) = -\alpha \Rightarrow \alpha(y + 1) = -py \Rightarrow \alpha = \frac{-py}{1 + y} \text{ satisfies } f(x) = 0.$$

$\therefore$ Substitute $x = \frac{-py}{1 + y}$ in $f(x) = 0$ and multiply by $(1 + y)^3$:

$$\begin{aligned} &(-py)^3 + p(-py)^2(1 + y) + q(-py)(1 + y)^2 + r(1 + y)^3 = 0 \\ &\Rightarrow -p^3y^3 + p^3y^2 + p^3y^3 - pqy(1 + 2y + y^2) + r(1 + 3y + 3y^2 + y^3) = 0 \\ &\Rightarrow \boxed{(r - pq)y^3 + (p^3 - 2pq + 3r)y^2 + (3r - pq)y + r = 0} \end{aligned}$$

is the required equation. ■

Problem 5.8

If α, β, γ are the roots of the equation $x^3 + 5x + 8 = 0$, form the new equation whose roots are $(\alpha + \beta)(\alpha + \gamma)$, $(\beta + \gamma)(\beta + \alpha)$, $(\gamma + \alpha)(\gamma + \beta)$.

Solution. Given $f(x) = x^3 + 5x + 8 = 0$ has α, β, γ as its roots

$$\Rightarrow S_1 = \alpha + \beta + \gamma = \frac{(-1)^1 0}{1} = 0, \quad S_2 = \alpha\beta + \beta\gamma + \gamma\alpha = 5, \quad S_3 = \alpha\beta\gamma = -8.$$

We have to form a new equation $g(y) = 0$ whose roots are $(\alpha + \beta)(\alpha + \gamma)$, $(\beta + \gamma)(\beta + \alpha)$, $(\gamma + \alpha)(\gamma + \beta)$.

$$\begin{aligned} \Rightarrow y = (\alpha + \beta)(\alpha + \gamma) \text{ satisfies } g(y) = 0, \quad y = (-\gamma)(-\beta) \quad (\because \text{from } S_1) \\ = \beta\gamma = -\frac{8}{\alpha} \quad (\because \text{from } S_3) \end{aligned}$$

$\Rightarrow \alpha = -\frac{8}{y}$ satisfies $f(x) = 0$. $\therefore$ Substitute $x = -\frac{8}{y}$ in $f(x) = 0$:

$$\left(-\frac{8}{y}\right)^3 + 5\left(-\frac{8}{y}\right) + 8 = 0 \Rightarrow -\frac{512}{y^3} - \frac{40}{y} + 8 = 0 \Rightarrow -\frac{64}{y^3} - \frac{5}{y} + 1 = 0 \Rightarrow -64 - 5y^2 + y^3 = 0.$$

Required equation is $y^3 - 5y^2 - 64 = 0$; it has $(\alpha + \beta)(\alpha + \gamma)$, $(\beta + \gamma)(\beta + \alpha)$, $(\gamma + \alpha)(\gamma + \beta)$ as its roots. ■

Problem 5.9

If α, β, γ are the roots of $x^3 - x - 1 = 0$, show that $\frac{1+\alpha}{1-\alpha} + \frac{1+\beta}{1-\beta} + \frac{1+\gamma}{1-\gamma} = -7$.

Solution. Given $f(x) = x^3 - x - 1 = 0$ has α, β, γ as its roots. Then $S_1 = \alpha + \beta + \gamma = 0$, $S_2 = \alpha\beta + \beta\gamma + \gamma\alpha = -1$, $S_3 = \alpha\beta\gamma = 1$.

To prove $\frac{1+\alpha}{1-\alpha} + \frac{1+\beta}{1-\beta} + \frac{1+\gamma}{1-\gamma} = -7$, let us form a new equation $g(y) = 0$ whose roots are $\frac{1+\alpha}{1-\alpha}$, $\frac{1+\beta}{1-\beta}$ and $\frac{1+\gamma}{1-\gamma}$.
 $\therefore y = \frac{1+\alpha}{1-\alpha}$ satisfies $g(y) = 0$. (It is already in single-root form, so there is no need to use S_1, S_2, S_3 .)

$$\Rightarrow (1 - \alpha)y = 1 + \alpha \Rightarrow \alpha(1 + y) = y - 1 \Rightarrow \alpha = \frac{y-1}{y+1} \text{ satisfies } f(x) = 0.$$

$\therefore$ Substitute $x = \frac{y-1}{y+1}$ in $f(x) = 0$:

$$\begin{aligned} \left(\frac{y-1}{y+1}\right)^3 - \left(\frac{y-1}{y+1}\right) - 1 &= 0 \Rightarrow (y-1)^3 - (y-1)(y+1)^2 - (y+1)^3 = 0 \\ \Rightarrow y^3 - 3y^2 + 3y - 1 - (y-1)(y^2 + 2y + 1) - (y^3 + 3y^2 + 3y + 1) &= 0 \\ \Rightarrow y^3 - 3y^2 + 3y - 1 - (y^3 + 2y^2 + y - y^2 - 2y - 1) - y^3 - 3y^2 - 3y - 1 &= 0 \\ \Rightarrow y^3(1 - 1 - 1) + y^2(-3 - 1 - 3) + y(3 + 1 - 3) + (-1 + 1 - 1) &= 0 \\ \Rightarrow -y^3 - 7y^2 + y - 1 = 0, \quad \text{i.e. } y^3 + 7y^2 - y + 1 = 0 \end{aligned}$$

is the required equation whose roots are $\frac{1+\alpha}{1-\alpha}$, $\frac{1+\beta}{1-\beta}$, $\frac{1+\gamma}{1-\gamma}$. Hence

$$S_1 = \frac{1+\alpha}{1-\alpha} + \frac{1+\beta}{1-\beta} + \frac{1+\gamma}{1-\gamma} = \frac{(-1)^1 7}{1} \Rightarrow \boxed{\frac{1+\alpha}{1-\alpha} + \frac{1+\beta}{1-\beta} + \frac{1+\gamma}{1-\gamma} = -7}$$

■

Exercise. If a, b, c are the roots of the equation $x^3 + px^2 + qx + r = 0$, find the equation whose roots are ab, bc, ca .
Ans: $y^3 - qy^2 + pry - r^2 = 0$.

6 Symmetric functions of roots

Generally for a cubic equation $x^3 + ax^2 + bx + c = 0$, if α, β, γ are the roots then we have

$$S_1 = \alpha + \beta + \gamma = -a; \quad S_2 = \alpha\beta + \beta\gamma + \gamma\alpha = b; \quad S_3 = \alpha\beta\gamma = -c.$$

Now we can write $S_1 = \alpha + \beta + \gamma = \sum \alpha$, $S_2 = \alpha\beta + \beta\gamma + \gamma\alpha = \sum \alpha\beta$.

Definition 6.1 (Symmetric function)

This kind of representation, in which the expression is unchanged by any interchange of the roots, is called a **symmetric function** of the roots. E.g.

$$\sum \alpha^2 = \alpha^2 + \beta^2 + \gamma^2, \quad \sum \alpha^2\beta = \alpha^2\beta + \beta^2\gamma + \gamma^2\alpha + \alpha^2\gamma + \beta^2\alpha + \gamma^2\beta.$$

6.1 Standard results on symmetric functions of the roots of a cubic

Result 6.1

- (1) $\sum \alpha^2 = (\sum \alpha)^2 - 2(\sum \alpha\beta) \quad (= S_1^2 - 2S_2)$
- (2) $\sum \alpha^2\beta = (\sum \alpha)(\sum \alpha\beta) - 3\alpha\beta\gamma \quad (= S_1S_2 - 3S_3)$
- (3) $\sum \alpha^2\beta^2 = (\sum \alpha\beta)^2 - 2(\sum \alpha)(\alpha\beta\gamma) \quad (= S_2^2 - 2S_1S_3)$
- (4) $\sum \alpha^3 = (\sum \alpha)^3 - 3(\sum \alpha)(\sum \alpha\beta) + 3\alpha\beta\gamma \quad (= S_1^3 - 3S_1S_2 + 3S_3)$

Proof of (1). Consider

$$\begin{aligned} S_1^2 &= (\sum \alpha)^2 = (\alpha + \beta + \gamma)^2 = \alpha^2 + \beta^2 + \gamma^2 + 2\alpha\beta + 2\beta\gamma + 2\gamma\alpha \\ &= \sum \alpha^2 + 2(\alpha\beta + \beta\gamma + \gamma\alpha) \\ (\sum \alpha)^2 &= \sum \alpha^2 + 2(\sum \alpha\beta) \Rightarrow \boxed{\sum \alpha^2 = (\sum \alpha)^2 - 2(\sum \alpha\beta)} \end{aligned}$$

Proof of (2). Consider

$$\begin{aligned} S_1S_2 &= (\sum \alpha) \left(\sum \alpha\beta \right) = (\alpha + \beta + \gamma)(\alpha\beta + \beta\gamma + \gamma\alpha) \\ &= \alpha^2\beta + \alpha\beta\gamma + \alpha^2\gamma + \beta^2\alpha + \beta^2\gamma + \alpha\beta\gamma + \alpha\beta\gamma + \gamma^2\beta + \gamma^2\alpha \\ &= (\alpha^2\beta + \alpha^2\gamma + \beta^2\alpha + \beta^2\gamma + \gamma^2\alpha + \gamma^2\beta) + 3(\alpha\beta\gamma) \\ (\sum \alpha) \left(\sum \alpha\beta \right) &= \sum \alpha^2\beta + 3\alpha\beta\gamma \Rightarrow \boxed{\sum \alpha^2\beta = (\sum \alpha)(\sum \alpha\beta) - 3\alpha\beta\gamma} \end{aligned}$$

(3) and (4) are checked similarly by expanding $S_2^2 = (\alpha\beta + \beta\gamma + \gamma\alpha)^2$ and $S_1^3 = (\alpha + \beta + \gamma)^3 = (\alpha + \beta + \gamma)(\alpha + \beta + \gamma)^2$ (exercise).

6.2 Problems

Problem 6.1

If α, β, γ are the roots of $x^3 + ax - b = 0$, find (i) $\sum \left(\frac{\alpha}{\beta\gamma} \right)$ (ii) $\sum \left(\frac{\alpha\beta}{\gamma} \right)$.

Solution. Given $f(x) = x^3 + ax - b = 0$ has α, β, γ as its roots $\Rightarrow \boxed{S_1 = \sum \alpha = 0}, \boxed{S_2 = \sum \alpha\beta = a}, \boxed{S_3 = \alpha\beta\gamma = b}$.

(i)

$$\sum \left(\frac{\alpha}{\beta\gamma} \right) = \frac{\alpha}{\beta\gamma} + \frac{\beta}{\gamma\alpha} + \frac{\gamma}{\alpha\beta}$$

$$\begin{aligned}
&= \frac{\alpha^2 + \beta^2 + \gamma^2}{\alpha\beta\gamma} = \frac{\sum \alpha^2}{b} \quad (\because \text{by } S_3) \\
&= \frac{(\sum \alpha)^2 - 2\sum \alpha\beta}{b} = \frac{S_1^2 - 2S_2}{b} = \frac{0^2 - 2(a)}{b} \Rightarrow \boxed{\sum \left(\frac{\alpha}{\beta\gamma} \right) = -\frac{2a}{b}}
\end{aligned}$$

(ii)

$$\begin{aligned}
\sum \left(\frac{\alpha\beta}{\gamma} \right) &= \frac{\alpha\beta}{\gamma} + \frac{\beta\gamma}{\alpha} + \frac{\gamma\alpha}{\beta} \\
&= \frac{(\alpha\beta)^2 + (\beta\gamma)^2 + (\gamma\alpha)^2}{\alpha\beta\gamma} = \frac{\sum \alpha^2 \beta^2}{\alpha\beta\gamma} \\
&= \frac{(\sum \alpha\beta)^2 - 2(\sum \alpha)(\alpha\beta\gamma)}{\alpha\beta\gamma} \quad (\because \text{by result (3)}) \\
&= \frac{S_2^2 - 2S_1S_3}{S_3} = \frac{a^2 - 0}{b} \Rightarrow \boxed{\sum \left(\frac{\alpha\beta}{\gamma} \right) = \frac{a^2}{b}}
\end{aligned}$$

■

Problem 6.2

Prove that the sum of the cubes of the roots of $x^3 - 6x^2 + 11x - 6 = 0$ is 36.

Solution. Given $f(x) = x^3 - 6x^2 + 11x - 6 = 0$ and let α, β, γ be the roots of $f(x) = 0$. Then

$$\boxed{S_1 = \sum \alpha = 6}; \quad \boxed{S_2 = \sum \alpha\beta = 11}; \quad \boxed{S_3 = \alpha\beta\gamma = 6}.$$

Now the sum of the cubes of the roots of $f(x) = 0$ is

$$\begin{aligned}
\alpha^3 + \beta^3 + \gamma^3 &= \sum \alpha^3 = (\sum \alpha)^3 - 3\left(\sum \alpha\right)\left(\sum \alpha\beta\right) + 3\alpha\beta\gamma \quad (\because \text{by result (4)}) \\
&= S_1^3 - 3S_1S_2 + 3S_3 = (6)^3 - 3(6)(11) + 3(6) = 216 - 198 + 18 = 36.
\end{aligned}$$

■

Problem 6.3

If α, β, γ are the roots of $x^3 - 14x + 8 = 0$, find (i) $\sum \alpha^2$ (ii) $\sum \alpha^3$.

Solution. Here $S_1 = 0, S_2 = -14, S_3 = -8$.

(i) $\sum \alpha^2 = S_1^2 - 2S_2 = 0 + 28 = \boxed{28}$.

(ii) $\sum \alpha^3 = S_1^3 - 3S_1S_2 + 3S_3 = 0 - 0 + 3(-8) = \boxed{-24}$.

■

Problem 6.4

If α, β, γ are the roots of $x^3 - 3ax + b = 0$, show that $\sum(\alpha - \beta)(\alpha - \gamma) = 9a$.

Solution. Here $S_1 = \sum \alpha = 0, S_2 = \sum \alpha\beta = -3a, S_3 = -b$.

$$\begin{aligned}
\sum(\alpha - \beta)(\alpha - \gamma) &= \sum (\alpha^2 - \alpha\gamma - \beta\alpha + \beta\gamma) \\
&= \sum \alpha^2 - \sum \alpha\gamma - \sum \alpha\beta + \sum \beta\gamma \\
&= \sum \alpha^2 - \sum \alpha\beta \quad (\text{each of } \sum \alpha\gamma, \sum \alpha\beta, \sum \beta\gamma \text{ equals } S_2) \\
&= (S_1^2 - 2S_2) - S_2 = S_1^2 - 3S_2 = 0 - 3(-3a) = 9a.
\end{aligned}$$

■

7 Additional problems in the solution of algebraic equations

Problem 7.1

Solve $2x^3 - x^2 - 22x - 24 = 0$, given that two of its roots are in the ratio 3 : 4.

Solution. Given $f(x) = 2x^3 - x^2 - 22x - 24 = 0$. Let α, β, γ be the roots of $f(x) = 0$. Given that two of its roots are in the ratio 3 : 4 $\Rightarrow$ the roots are $3\alpha, 4\alpha, \gamma$.

$$S_1 = 3\alpha + 4\alpha + \gamma = \frac{(-1)^1(-1)}{2} \Rightarrow 7\alpha + \gamma = \frac{1}{2} \Rightarrow \gamma = \frac{1}{2} - 7\alpha.$$

Now

$$\begin{aligned} S_2 &= (3\alpha)(4\alpha) + (4\alpha)\gamma + \gamma(3\alpha) = \frac{-22}{2} = -11 \\ &\Rightarrow 12\alpha^2 + 7\alpha\gamma = -11 \\ &\Rightarrow 12\alpha^2 + 7\alpha\left(\frac{1}{2} - 7\alpha\right) = -11 \quad (\because \text{from } S_1) \\ &\Rightarrow 12\alpha^2 + \frac{7}{2}\alpha - 49\alpha^2 = -11 \\ &\Rightarrow -37\alpha^2 + \frac{7}{2}\alpha + 11 = 0 \Rightarrow 74\alpha^2 - 7\alpha - 22 = 0 \\ &\Rightarrow (2\alpha + 1)(37\alpha - 22) = 0 \Rightarrow \alpha = -\frac{1}{2} \text{ or } \frac{22}{37}. \end{aligned}$$

If $\alpha = -\frac{1}{2}$, then the roots $3\alpha, 4\alpha, \frac{1}{2} - 7\alpha$ are $-\frac{3}{2}, -2, 4$. When $x = -2$,

$$f(-2) = 2(-2)^3 - (-2)^2 - 22(-2) - 24 = -16 - 4 + 44 - 24 = 0,$$

satisfies the equation. $\therefore$ Roots are $\boxed{-\frac{3}{2}, -2, 4}$. (Definitely, when $\alpha = \frac{22}{37}$, the respective roots will not satisfy $f(x) = 0$, since then $S_3 = 12\alpha^2\gamma \neq 12$.) ■

Problem 7.2

Solve $x^4 - 2x^3 + 4x^2 + 6x - 21 = 0$, given that two of its roots are equal in magnitude but opposite in sign.

Solution. Given $f(x) = x^4 - 2x^3 + 4x^2 + 6x - 21 = 0$; let $\alpha, \beta, \gamma, \delta$ be the roots of $f(x) = 0$. Given that two of its roots are equal in magnitude and opposite in sign $\Rightarrow \alpha = -\beta \Rightarrow \boxed{\alpha + \beta = 0}$. Now $S_1 = \alpha + \beta + \gamma + \delta = 2 \Rightarrow \boxed{\gamma + \delta = 2}$. Now

$$\begin{aligned} f(x) &= (x - \alpha)(x - \beta)(x - \gamma)(x - \delta) = (x^2 - (\alpha + \beta)x + \alpha\beta)(x^2 - (\gamma + \delta)x + \gamma\delta) \\ &= (x^2 - 0x + \alpha\beta)(x^2 - 2x + \gamma\delta) = (x^2 + \lambda)(x^2 - 2x + \mu), \quad \text{where } \lambda = \alpha\beta, \mu = \gamma\delta \\ &= x^4 - 2x^3 + \mu x^2 + \lambda x^2 - 2\lambda x + \lambda\mu. \end{aligned}$$

$\therefore x^4 - 2x^3 + 4x^2 + 6x - 21 = x^4 - 2x^3 + (\mu + \lambda)x^2 - 2\lambda x + \lambda\mu$. Comparing: $\mu + \lambda = 4$; $-2\lambda = 6 \Rightarrow \boxed{\lambda = -3}$, $\boxed{\mu = 7}$ (check: $\lambda\mu = -21$ ✓).

$$\therefore f(x) = (x^2 + \lambda)(x^2 - 2x + \mu) = (x^2 - 3)(x^2 - 2x + 7).$$

$\Rightarrow$ Roots of $f(x) = 0$ are the roots of $x^2 - 3 = 0$ and $x^2 - 2x + 7 = 0 \Rightarrow x = \pm\sqrt{3}$; $x = \frac{2 \pm \sqrt{4 - 28}}{2} = 1 \pm i\sqrt{6}$.

$\Rightarrow$ Roots of $f(x) = 0$ are $\boxed{\pm\sqrt{3}, 1 \pm i\sqrt{6}}$. ■

Problem 7.3

Solve $6x^4 - 11x^3 + 11x^2 - 5x + 2 = 0$ if the product of two roots is unity.

Solution. Let $\alpha, \beta, \gamma, \delta$ be the roots; given $\boxed{\alpha\beta = 1}$. $\Rightarrow S_4 = \alpha\beta\gamma\delta = \frac{(-1)^4 2}{6} = \frac{1}{3} \Rightarrow \boxed{\gamma\delta = \frac{1}{3}}$.

$$f(x) = 6(x - \alpha)(x - \beta)(x - \gamma)(x - \delta) = 6(x^2 - (\alpha + \beta)x + \alpha\beta)(x^2 - (\gamma + \delta)x + \gamma\delta)$$

$$6x^4 - 11x^3 + 11x^2 - 5x + 2 = 6(x^2 - \lambda x + 1)(x^2 - \mu x + \frac{1}{3}), \quad \lambda = \alpha + \beta, \mu = \gamma + \delta$$

$$= 6x^4 - 6(\lambda + \mu)x^3 + 6(\frac{4}{3} + \lambda\mu)x^2 - 6(\frac{\lambda}{3} + \mu)x + 2.$$

Comparing: $6(\lambda + \mu) = 11$, $6(\frac{4}{3} + \lambda\mu) = 11 \Rightarrow \lambda\mu = \frac{1}{2}$, and $2\lambda + 6\mu = 5$. From $2\lambda + 2\mu = \frac{11}{3}$ and $2\lambda + 6\mu = 5$:

$$4\mu = \frac{4}{3} \Rightarrow \mu = \frac{1}{3}, \lambda = \frac{3}{2} \quad (\text{check: } \lambda\mu = \frac{1}{2} \checkmark).$$

$$\therefore f(x) = 6(x^2 - \frac{3}{2}x + 1)(x^2 - \frac{1}{3}x + \frac{1}{3}) = (2x^2 - 3x + 2)(3x^2 - x + 1).$$

$$2x^2 - 3x + 2 = 0 \Rightarrow x = \frac{3 \pm i\sqrt{7}}{4}; \quad 3x^2 - x + 1 = 0 \Rightarrow x = \frac{1 \pm i\sqrt{11}}{6}. \therefore \text{Roots are } \frac{3 \pm i\sqrt{7}}{4}, \frac{1 \pm i\sqrt{11}}{6}. \quad \blacksquare$$

7.1 Multiple roots

Definition 7.1 (Multiple roots)

If an equation $f(x) = 0$ has roots such as $\alpha, \alpha, \beta, \beta, \gamma, \delta, \dots$, then α and β are called **multiple roots** and they have multiplicity 2.

Result 7.1 (Remark)

If $f(x) = 0$ has a multiple root α of multiplicity r , then $f'(x) = 0$ has the multiple root α with multiplicity $(r - 1)$, $f''(x) = 0$ has the multiple root α with multiplicity $(r - 2)$, and so on.

Problem 7.4

Solve the equation $x^3 - 4x^2 + 5x - 2 = 0$, given that it has a double root (i.e. a root appears two times).

Solution. Given $f(x) = x^3 - 4x^2 + 5x - 2 = 0$ has a double root. $\therefore$ The roots may be α, α, β . $\Rightarrow f'(x) = 3x^2 - 8x + 5 = 0$ has the roots α, γ . Roots of $f'(x) = 0$: $3x^2 - 8x + 5 = 0 \Rightarrow 3x(x - 1) - 5(x - 1) = 0 \Rightarrow x = 1, \frac{5}{3}$, i.e. $x = 1$ and $x = \frac{5}{3}$ are the roots of $f'(x) = 0$. Now substitute $x = 1$ in $f(x)$: $1 - 4 + 5 - 2 = 0 \Rightarrow x = 1$ is the double root of $f(x) = 0 \Rightarrow$ roots of $f(x) = 0$ are $1, 1, \beta$. $\therefore$ Product of the roots $= 1 \cdot 1 \cdot \beta = \frac{(-1)^3(-2)}{1} \Rightarrow \boxed{\beta = 2}$. $\therefore$ Roots of $f(x) = 0$ are $\boxed{1, 1, 2}$. $\blacksquare$

Problem 7.5

Solve $8x^4 + 4x^3 - 18x^2 + 11x - 2 = 0$, given that it has three equal roots.

Solution. Given $f(x) = 8x^4 + 4x^3 - 18x^2 + 11x - 2 = 0$ has three equal roots; let them be $\alpha, \alpha, \alpha, \beta$. $\Rightarrow f'(x) = 32x^3 + 12x^2 - 36x + 11 = 0$ has the roots $\alpha, \alpha, \gamma \Rightarrow f''(x) = 96x^2 + 24x - 36 = 0$ has the roots α, δ , i.e.

$$f''(x) = 0 \Rightarrow 8x^2 + 2x - 3 = 0 \Rightarrow 8x^2 - 4x + 6x - 3 = 0 \Rightarrow 4x(2x - 1) + 3(2x - 1) = 0 \Rightarrow \boxed{x = \frac{1}{2}, -\frac{3}{4}}$$

Now substitute $x = \frac{1}{2}$ in $f(x)$:

$$8\left(\frac{1}{16}\right) + 4\left(\frac{1}{8}\right) - 18\left(\frac{1}{4}\right) + \frac{11}{2} - 2 = \frac{1}{2} + \frac{1}{2} - \frac{9}{2} + \frac{11}{2} - 2 = 0,$$

i.e. $x = \frac{1}{2}$ satisfies both $f''(x)$ and $f(x) \Rightarrow x = \frac{1}{2}$ is the multiple root of $f(x) = 0$. $\Rightarrow$ Roots of $f(x) = 0$ are $\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \beta$. $\therefore$ Product of the roots $= \left(\frac{1}{2}\right)^3 \beta = \frac{(-1)^4(-2)}{8} \Rightarrow \frac{\beta}{8} = -\frac{2}{8} \Rightarrow \boxed{\beta = -2}$. $\therefore$ Roots of $f(x) = 0$ are $\boxed{\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -2}$. $\blacksquare$

Problem 7.6

Solve $x^4 + 4x^3 - 2x^2 - 12x + 9 = 0$, given that it has two pairs of equal roots.

Solution. Given $f(x) = x^4 + 4x^3 - 2x^2 - 12x + 9 = 0$ has two pairs of equal roots; let them be $\alpha, \alpha, \beta, \beta$. $\Rightarrow$ Product of the roots $= \alpha^2 \beta^2 = \frac{(-1)^4 9}{1} \Rightarrow (\alpha\beta)^2 = 9 \Rightarrow \alpha\beta = \pm 3$.
 $\Rightarrow f'(x) = 4x^3 + 12x^2 - 4x - 12 = 0$ has the roots α, β, γ . Now

$$f'(x) = 4(x^3 + 3x^2 - x - 3) = 4(x^2(x+3) - (x+3)) = 4(x+3)(x-1)(x+1),$$

so the roots of $f'(x) = 0$ are $1, -1, -3$. The repeated roots of f must be among these and must satisfy $f(x) = 0$:

$$f(1) = 1 + 4 - 2 - 12 + 9 = 0, \quad f(-3) = 81 - 108 - 18 + 36 + 9 = 0, \quad f(-1) = 1 - 4 - 2 + 12 + 9 = 16 \neq 0.$$

$\therefore \alpha = 1, \beta = -3$ (and $\alpha\beta = -3$, consistent with $(\alpha\beta)^2 = 9$). Indeed $f(x) = (x^2 + 2x - 3)^2 = (x-1)^2(x+3)^2$. $\therefore$ Roots of $f(x) = 0$ are $\boxed{1, 1, -3, -3}$. ■

Problem 7.7

Find the value of k so that the equation $2x^3 - 9x^2 + 12x + k = 0$ may have a double root.

Solution. Given $f(x) = 2x^3 - 9x^2 + 12x + k = 0$ may have a double root; let the roots be α, α, β . $\therefore f'(x) = 6x^2 - 18x + 12 = 0$ may have the roots $\alpha, \gamma \Rightarrow x^2 - 3x + 2 = 0 \Rightarrow$ roots of $f'(x) = 0$ are $x = 1, 2$.

Suppose $x = 1$ is a double root of $f(x) = 0$; then $f(1) = 2 - 9 + 12 + k = 0 \Rightarrow \boxed{k = -5}$.

Suppose $x = 2$ is a double root of $f(x) = 0$; then $f(2) = 16 - 36 + 24 + k = 0 \Rightarrow \boxed{k = -4}$. ■

8 Descartes' rule of signs

Result 8.1 (Descartes' rule of signs)

- (1) An equation $f(x) = 0$ cannot have more positive roots than there are changes of sign in the terms of the polynomial $f(x)$.
- (2) An equation $f(x) = 0$ cannot have more negative roots than there are changes of sign in the terms of the polynomial $f(-x)$.

E.g. in $x^3 + x^2 + x - 1$ the signs are $+$ $+$ $+$ $-$: one change of sign.

Problem 8.1

Determine the nature of the roots of $x^6 + 3x^5 + 5x - 1 = 0$.

Solution. Given $f(x) = x^6 + 3x^5 + 5x - 1 = 0$. Here the signs are $+$ $+$ $+$ $-$, i.e. we have only one change of sign for the terms of $f(x) = 0 \Rightarrow$ at maximum level, $f(x) = 0$ can have one positive root.

Similarly for $f(-x) = (-x)^6 + 3(-x)^5 + 5(-x) - 1 = 0$, i.e. $f(-x) = x^6 - 3x^5 - 5x - 1 = 0$, the signs are $+$ $-$ $-$ $-$. Here we have only one change of sign for the terms of $f(-x) = 0 \Rightarrow$ at maximum, $f(x) = 0$ has one negative root.

I.e. we have (at most) one positive and one negative real root. $\therefore$ The remaining 4 roots must be complex roots (and $x = 0$ is not a root). ■

Exercise. Determine the nature of the roots of $x^5 - 6x^2 - 4x + 7 = 0$.

9 Stability analysis of control systems (Routh–Hurwitz criterion)

In electrical engineering, the stability of a control system can be determined by the closed loop transfer function

$$M(s) = \frac{b_0 s^m + b_1 s^{m-1} + \dots + b_m}{a_0 s^n + a_1 s^{n-1} + \dots + a_n},$$

in which the denominator polynomial $a_0s^n + a_1s^{n-1} + \dots + a_n$ is called the **characteristic polynomial** of the control system. This $M(s)$ is obtained by transferring a time domain function $f(t)$ to a frequency domain function $M(s)$ with the help of the Laplace transform (you can study this in detail in later semesters).

Here we can determine the stability of the control system using the nature of the roots of the characteristic equation (C.E.) $a_0s^n + a_1s^{n-1} + \dots + a_n = 0$. But when the degree of the C.E. is higher, finding the roots is somewhat difficult; hence we have the alternative method called the **Routh–Hurwitz stability criterion**. In this criterion we have two conditions: one is the necessary condition and the other one is the sufficient condition.

Result 9.1 (Routh–Hurwitz criterion)

Necessary condition: the coefficients of the C.E. should be positive, i.e. all coefficients must be non-zero and positive.

Sufficient condition: all the elements of the first column of the Routh array should have the same sign (either all +ve or all –ve) and there is no change of sign.

When these two conditions are satisfied, the system is **stable**.

9.1 Procedure involved in the Routh array method

Suppose we have the characteristic polynomial $a_0s^n + a_1s^{n-1} + a_2s^{n-2} + \dots + a_ns^0$. First let us construct the Routh array table as follows. Fill up the first two rows using the coefficients of the characteristic polynomial:

s^n	a_0	a_2	a_4	a_6	$\dots$
s^{n-1}	a_1	a_3	a_5	a_7	$\dots$
s^{n-2}	$b_1 = \frac{a_1a_2 - a_0a_3}{a_1}$	$b_2 = \frac{a_1a_4 - a_0a_5}{a_1}$	$b_3 = \frac{a_1a_6 - a_0a_7}{a_1}$	$\dots$	
s^{n-3}	$c_1 = \frac{b_1a_3 - a_1b_2}{b_1}$	$c_2 = \frac{b_1a_5 - a_1b_3}{b_1}$	$\dots$		
s^{n-4}	$d_1 = \frac{c_1b_2 - b_1c_2}{c_1}$	$\dots$			
$\vdots$	$\vdots$				
s^0					

- In the construction of the Routh array, if any row's elements have a common factor we can cancel the common factor and write it in a simplified form, i.e. suppose we have 2, 4, 8 as row elements; then we can write it as 1, 2, 4.
- Now consider the number of sign changes in the first column of the Routh table.
- If there is no change of sign in the first column of the Routh table, then the system is stable; if there is a change of sign, we can say that the system is unstable.

Problem 9.1

Determine the stability of the control system having the characteristic equation $s^4 + 3s^3 + 3s^2 + 2s + 1 = 0$.

Solution. Clearly all the coefficients of the C.E. $s^4 + 3s^3 + 3s^2 + 2s + 1 = 0$ are positive $\Rightarrow$ it satisfies the necessary condition. For the sufficient condition, let us construct the Routh array table as follows:

$$\begin{array}{c|ccc}
s^4 & 1 & 3 & 1 \\
s^3 & 3 & 2 & - \\
s^2 & \frac{3 \times 3 - 2 \times 1}{3} = \frac{7}{3} & \frac{3 \times 1 - 1 \times 0}{3} = 1 & - \\
s^1 & \frac{\frac{7}{3} \times 2 - 3(1)}{\frac{7}{3}} = \frac{5/3}{7/3} = \frac{5}{7} & \frac{\frac{7}{3}(0) - 3(0)}{7/3} = 0 & - \\
s^0 & \frac{\frac{5}{7}(1) - \frac{7}{3}(0)}{5/7} = 1 & &
\end{array}$$

Here there is no change of sign in the first column of the Routh table $\Rightarrow$ the given system is a **stable** one. ■

9.2 Special cases in the Routh array

For the following cases, construction of the Routh array elements is difficult: (i) the first element of any row of the Routh array is zero; (ii) all the elements of any row of the Routh array are zero.

Result 9.2 (Case (i): first element of a row is zero)

Suppose the first element of any row is zero, and at least one element is non-zero in that row. Replace the first zero element by a small number ϵ (≈ 0.0001), calculate the further row elements as usual and identify the number of sign changes in the first column of the Routh array, considering $\epsilon \rightarrow 0$.

Problem 9.2

Identify the stability of the control system having the C.E. $s^4 + 2s^3 + s^2 + 2s + 1 = 0$.

Solution. All the coefficients of the C.E. are positive; the necessary condition is satisfied. Let us construct the Routh array table:

$$\begin{array}{c|ccc}
s^4 & 1 & 1 & 1 \\
s^3 & 2 & 2 & - \\
s^2 & \frac{2 \times 1 - 2 \times 1}{2} = 0 \rightarrow \epsilon & \frac{2 \times 1 - 1 \times 0}{2} = 1 & - \\
s^1 & \frac{2\epsilon - 2}{\epsilon} & \frac{\epsilon(0) - 2(0)}{\epsilon} = 0 & - \\
s^0 & 1 & &
\end{array}$$

Consider the number of sign changes in the first column by treating $\epsilon \rightarrow 0$. Then the first column becomes 1, 2, 0^+ , $-\infty$, 1, i.e. we have two changes of sign in the first column $\Rightarrow$ the system is **unstable**. ■

Result 9.3 (Case (ii): all the elements of a row are zero)

In this case we can proceed in the following way:

- (1) Construct the auxiliary equation $A(s)$ from the elements just above the row where all elements are zero.
- (2) Identify $\frac{dA}{ds}$ and substitute the coefficients of $\frac{dA}{ds}$ as the elements of the row in which all the elements are zero.
- (3) Proceed further as usual.

Problem 9.3

Identify the stability of the control system having C.E. $s^5 + 3s^4 + s^3 + 3s^2 + s + 3 = 0$.

Solution. All the coefficients of the C.E. are positive $\Rightarrow$ the necessary condition is satisfied. Let us construct the Routh array table:

$$\begin{array}{c|cccc} s^5 & 1 & & 1 & \\ s^4 & \cancel{3} 1 & \cancel{3} 1 & \cancel{3} 1 & (\because \text{common multiple 3 cancelled}) \\ s^3 & \frac{1-1}{1} = 0 & \frac{1-1}{1} = 0 & - & (\text{all the elements in the row are zero}) \end{array}$$

$\therefore$ Construct the auxiliary equation (A.E.) from the s^4 row: $A(s) = s^4 + s^2 + 1 \Rightarrow \frac{dA}{ds} = 4s^3 + 2s$; substitute these coefficients as the elements of the s^3 row. Now we have the following table:

$$\begin{array}{c|cccc} s^5 & 1 & & 1 & \\ s^4 & 1 & & 1 & \\ s^3 & \cancel{4} 2 & \cancel{2} 1 & - & (\because \text{common multiple cancelled}) \\ s^2 & \frac{2-1}{2} = \frac{1}{2} & \frac{2-0}{2} = 1 & & - \\ s^1 & \frac{\frac{1}{2}-2}{1/2} = -3 & 0 & & - \\ s^0 & \frac{-3-0}{-3} = 1 & & & \end{array}$$

We can see from the first column of the Routh table there are two changes of sign, i.e. $(\frac{1}{2} \rightarrow -3)$ and $(-3 \rightarrow 1)$. $\therefore$ The given system is **unstable**. ■

Problem 9.4

Discuss the stability of the system determined by the C.E. $s^6 + 3s^5 + 4s^4 + 6s^3 + 5s^2 + 3s + 2 = 0$.

Solution. The given C.E. has all the coefficients positive $\Rightarrow$ the necessary condition is satisfied. Let us construct the Routh table as follows:

$$\begin{array}{c|cccc} s^6 & 1 & 4 & 5 & 2 \\ s^5 & 3 & 6 & 3 & - \\ s^4 & \cancel{2} 1 & \cancel{4} 2 & \cancel{2} 1 & - (\because \text{common multiplier 2 cancelled}) \\ s^3 & 0 & 0 & - & - (\text{elements are zero}) \end{array}$$

(Here the s^5 row simplifies to 1, 2, 1 and then the s^4 row is $\frac{1 \cdot 4 - 1 \cdot 2}{1} = 2$, $\frac{1 \cdot 5 - 1 \cdot 1}{1} = 4$, 2, i.e. 1, 2, 1.) $\Rightarrow$ A.E. is $A(s) = s^4 + 2s^2 + 1 \Rightarrow \frac{dA}{ds} = 4s^3 + 4s$; substitute in the s^3 row:

s^6	1	4	5		2
s^5	3	6	3		–
s^4	1	2	1		–
s^3	1	1	–	–	(new row inserted)
s^2	1	1	–		–
s^1	0	–	–	–	(row with zero elements)
s^1	2	–	–	–	(A.E. $A(s) = s^2 + 1 \Rightarrow \frac{dA}{ds} = 2s$; new row inserted)
s^0	1	–	–		–

$\Rightarrow$ There is no change of sign in the first column of the Routh array $\Rightarrow$ the given system is stable. ■

Note

Whenever a whole row of zeros occurs, the auxiliary polynomial ($s^2 + 1$ here) is a factor of the C.E.; its roots $s = \pm i$ lie on the imaginary axis. With no sign change and such roots, the system is said to be *marginally stable* (oscillatory but not growing).

Exercises.

- (3) Identify the stability of the system given by the C.E. $s^5 - 2s^4 - 10s^3 + 8s^2 + 33s + 18 = 0$.

Ans: The necessary condition is not satisfied $\therefore$ the system is unstable.

- (4) C.E. is $s^5 + 4s^4 + s^3 + s^2 + 4s + 1 = 0$.

Ans: The system is unstable, because of changes of sign in the first column of the Routh table.

Result 9.4 (Remark: locating the roots)

- In the first column of the Routh array table, if all the elements are positive (there is no change of sign), then all the roots of the C.E. lie in the left half of the s -plane (xy -plane), i.e. all the roots have negative real parts. (This is the condition for a stable system.)
- If there is a change of sign in the first column of the Routh table, we can say that the number of roots lying in the right half of the s -plane (roots with positive real parts) is as many as the number of changes of sign, i.e. one change of sign $\Rightarrow$ one root in the right half of the s -plane; two changes of sign $\Rightarrow$ two roots in the right half of the s -plane.

In this way we can locate the roots.

10 Algebra: Binomial series

By the Binomial theorem, we have

$$(x + a)^n = x^n + {}^nC_1 x^{n-1} a + {}^nC_2 x^{n-2} a^2 + \cdots + {}^nC_r x^{n-r} a^r + \cdots + a^n,$$

where

$${}^nC_r = \frac{n!}{(n-r)! r!} = \frac{n(n-1)(n-2) \cdots (n-(r-1))(n-r)!}{(n-r)! r!} \Rightarrow {}^nC_r = \frac{n(n-1)(n-2) \cdots (n-r+1)}{r!},$$

i.e. ${}^nC_1 = n$, ${}^nC_2 = \frac{n(n-1)}{2!}$, ${}^nC_3 = \frac{n(n-1)(n-2)}{3!}$, ${}^nC_4 = \frac{n(n-1)(n-2)(n-3)}{4!}$. Hence

$$(x + a)^n = x^n + nx^{n-1}a + \frac{n(n-1)}{2!}x^{n-2}a^2 + \frac{n(n-1)(n-2)}{3!}x^{n-3}a^3 + \cdots + a^n.$$

From the above Binomial theorem we have the following series for $-1 < x < 1$ and for any value of n (rational):

Result 10.1 (Binomial series, $|x| < 1$)

$$(1) \quad (1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots$$

$$(2) \quad (1-x)^n = 1 - nx + \frac{n(n-1)}{2!}x^2 - \frac{n(n-1)(n-2)}{3!}x^3 + \dots$$

$$(3) \quad (1+x)^{-n} = 1 - nx + \frac{n(n+1)}{2!}x^2 - \frac{n(n+1)(n+2)}{3!}x^3 + \dots$$

$$(4) \quad (1-x)^{-n} = 1 + nx + \frac{n(n+1)}{2!}x^2 + \frac{n(n+1)(n+2)}{3!}x^3 + \dots$$

In the above series (3) and (4), if we place particular values for n such as $n = 1, 2, 3, \dots$, we have the following infinite series:

$$(i) \quad (1+x)^{-1} = 1 - x + x^2 - x^3 + x^4 - \dots$$

$$(ii) \quad (1-x)^{-1} = 1 + x + x^2 + x^3 + \dots$$

$$(iii) \quad (1+x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \dots$$

$$(iv) \quad (1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \dots$$

$$(v) \quad (1+x)^{-3} = 1 - 3x + 6x^2 - 10x^3 + \dots$$

$$(vi) \quad (1-x)^{-3} = 1 + 3x + 6x^2 + 10x^3 + \dots$$

Again from series (3) and (4), when n is rational such as $n = \frac{p}{q}$, we have the following infinite series:

$$(1+x)^{-p/q} = 1 - \frac{p}{q}x + \frac{\frac{p}{q}\left(\frac{p}{q}+1\right)}{2!}x^2 - \frac{\frac{p}{q}\left(\frac{p}{q}+1\right)\left(\frac{p}{q}+2\right)}{3!}x^3 + \dots = 1 - \frac{p}{q}x + \frac{p(p+q)}{2!q^2}x^2 - \frac{p(p+q)(p+2q)}{3!q^3}x^3 + \dots$$

Result 10.2 (Binomial series with rational index)

$$(1+x)^{-p/q} = 1 - \frac{p}{1!}\left(\frac{x}{q}\right) + \frac{p(p+q)}{2!}\left(\frac{x}{q}\right)^2 - \frac{p(p+q)(p+2q)}{3!}\left(\frac{x}{q}\right)^3 + \dots$$

$$(1-x)^{-p/q} = 1 + \frac{p}{1!}\left(\frac{x}{q}\right) + \frac{p(p+q)}{2!}\left(\frac{x}{q}\right)^2 + \frac{p(p+q)(p+2q)}{3!}\left(\frac{x}{q}\right)^3 + \dots$$

10.1 Problems**Problem 10.1**

If $|x| < \frac{1}{2}$, prove that the coefficient of x^n in the expansion of $(2-4x)(1-2x)^{-2}$ is 2^{n+1} .

Solution. Given

$$\begin{aligned} (2-4x)(1-2x)^{-2} &= 2(1-2x)(1-2x)^{-2} = 2(1-2x)^{-1} \\ &= 2\left(1 + (2x) + (2x)^2 + (2x)^3 + \dots + (2x)^n + \dots\right) \quad (\because (1-x)^{-1} = 1 + x + x^2 + x^3 + \dots) \\ &= 2\left[1 + 2x + 2^2x^2 + 2^3x^3 + \dots + 2^nx^n + \dots\right] \end{aligned}$$

$$\Rightarrow \text{Coefficient of } x^n = 2 \cdot 2^n \Rightarrow \boxed{\text{Coefficient of } x^n = 2^{n+1}}. \quad \blacksquare$$

Problem 10.2

Find the coefficient of x^n in the expansion of $(1-2x+3x^2-4x^3+\dots)^{-n}$.

Solution. Given $(1-2x+3x^2-4x^3+\dots)^{-n} = ((1+x)^{-2})^{-n} = (1+x)^{2n}$

$$= 1 + (2n)x + \frac{(2n)(2n-1)}{2!}x^2 + \frac{(2n)(2n-1)(2n-2)}{3!}x^3 + \dots + \frac{(2n)(2n-1)(2n-2)\dots(2n-(n-1))}{n!}x^n + \dots$$

$$\Rightarrow \boxed{\text{Coefficient of } x^n = \frac{2n(2n-1)(2n-2)\dots(n+1)}{n!}} \quad \left(= \frac{(2n)!}{(n!)^2}\right).$$

**Problem 10.3**

If x is small, prove that $\left(\frac{1-x}{1+x}\right)^{1/2} = 1 - x + \frac{x^2}{2} - \frac{x^3}{2}$ (approximately).

Solution. Given

$$\begin{aligned}
 \left(\frac{1-x}{1+x}\right)^{1/2} &= (1-x)^{1/2}(1+x)^{-1/2} \\
 &= \left[1 - \frac{1}{2}x + \frac{\frac{1}{2}(\frac{1}{2}-1)}{2!}x^2 - \frac{\frac{1}{2}(\frac{1}{2}-1)(\frac{1}{2}-2)}{3!}x^3 + \dots\right] \\
 &\quad \times \left[1 - \frac{1}{2}x + \frac{\frac{1}{2}(\frac{1}{2}+1)}{2!}x^2 - \frac{\frac{1}{2}(\frac{1}{2}+1)(\frac{1}{2}+2)}{3!}x^3 + \dots\right] \\
 &= \left[1 - \frac{x}{2} - \frac{x^2}{8} - \frac{3}{48}x^3\right] \left[1 - \frac{x}{2} + \frac{3}{8}x^2 - \frac{15}{48}x^3\right] \quad (\because x \text{ is small, neglect } x^4 \text{ and higher powers}) \\
 &= \left(1 - \frac{x}{2} - \frac{x^2}{8} - \frac{1}{16}x^3\right) \left(1 - \frac{x}{2} + \frac{3x^2}{8} - \frac{5}{16}x^3\right) \\
 &= 1 - \frac{x}{2} + \frac{3x^2}{8} - \frac{5}{16}x^3 - \frac{x}{2} + \frac{x^2}{4} - \frac{3x^3}{16} - \frac{x^2}{8} + \frac{x^3}{16} - \frac{1}{16}x^3 + (\text{higher powers}) \\
 &= 1 - x + x^2 \left(\frac{3}{8} + \frac{1}{4} - \frac{1}{8}\right) + x^3 \left(-\frac{5}{16} - \frac{3}{16} + \frac{1}{16} - \frac{1}{16}\right) \\
 &\Rightarrow \boxed{\left(\frac{1-x}{1+x}\right)^{1/2} = 1 - x + \frac{x^2}{2} - \frac{x^3}{2} \quad (\text{approx.})}
 \end{aligned}$$

**10.2 Summation of series using binomial series****Problem 10.4**

Sum to infinity the series $1 + \frac{1}{5} + \frac{1 \cdot 4}{5 \cdot 10} + \frac{1 \cdot 4 \cdot 7}{5 \cdot 10 \cdot 15} + \dots$

Solution. Let

$$\begin{aligned}
 S &= 1 + \frac{1}{5} + \frac{1 \cdot 4}{5 \cdot 10} + \frac{1 \cdot 4 \cdot 7}{5 \cdot 10 \cdot 15} + \dots \\
 &= 1 + \frac{1}{1 \cdot 5} + \frac{1 \cdot 4}{(1 \cdot 5)(2 \cdot 5)} + \frac{1 \cdot 4 \cdot 7}{(1 \cdot 5)(2 \cdot 5)(3 \cdot 5)} + \dots \\
 &= 1 + \frac{1}{1!5} + \frac{1 \cdot 4}{2!5^2} + \frac{1 \cdot 4 \cdot 7}{3!5^3} + \dots \\
 &= 1 + \frac{1}{1} \left(\frac{1}{5}\right) + \frac{1 \cdot 4}{2!} \left(\frac{1}{5}\right)^2 + \frac{1 \cdot 4 \cdot 7}{3!} \left(\frac{1}{5}\right)^3 + \dots
 \end{aligned}$$

is of the form

$$1 + \frac{p}{1!} \left(\frac{x}{q}\right) + \frac{p(p+q)}{2!} \left(\frac{x}{q}\right)^2 + \frac{p(p+q)(p+2q)}{3!} \left(\frac{x}{q}\right)^3 + \dots,$$

where $\boxed{p=1}$; $p+q=4 \Rightarrow \boxed{q=3}$; $\frac{x}{q} = \frac{1}{5} \Rightarrow \boxed{x = \frac{3}{5}}$.

$$\therefore S = (1-x)^{-p/q} = \left(1 - \frac{3}{5}\right)^{-1/3} = \left(\frac{2}{5}\right)^{-1/3} = \left(\frac{5}{2}\right)^{1/3} \Rightarrow \boxed{S = \left(\frac{5}{2}\right)^{1/3} = \sqrt[3]{5/2}}$$

Problem 10.5

Sum the series to ∞ : $\left(\frac{1}{2}\right)^2 + \frac{1}{2!} \left(\frac{1}{2}\right)^4 + \frac{1 \cdot 3}{3!} \left(\frac{1}{2}\right)^6 + \dots$

Solution. Let $S = \left(\frac{1}{2}\right)^2 + \frac{1}{2!} \left(\frac{1}{2}\right)^4 + \frac{1 \cdot 3}{3!} \left(\frac{1}{2}\right)^6 + \dots$. Multiplying by -1 (to create the factor $p = -1$):

$$\begin{aligned} -1 \cdot S &= \frac{-1}{1!} \left(\frac{1}{2}\right)^2 + \frac{-1 \cdot 1}{2!} \left(\frac{1}{2}\right)^4 + \frac{-1 \cdot 1 \cdot 3}{3!} \left(\frac{1}{2}\right)^6 + \dots \\ &= \frac{-1}{1!} \left(\left(\frac{1}{2}\right)^2\right)^1 + \frac{-1 \cdot 1}{2!} \left(\left(\frac{1}{2}\right)^2\right)^2 + \frac{-1 \cdot 1 \cdot 3}{3!} \left(\left(\frac{1}{2}\right)^2\right)^3 + \dots \\ -1 \cdot S &= \frac{-1}{1!} \left(\frac{1}{4}\right) + \frac{-1 \cdot 1}{2!} \left(\frac{1}{4}\right)^2 + \frac{-1 \cdot 1 \cdot 3}{3!} \left(\frac{1}{4}\right)^3 + \dots \\ \Rightarrow 1 - S &= 1 + \frac{-1}{1!} \left(\frac{1}{4}\right) + \frac{-1 \cdot 1}{2!} \left(\frac{1}{4}\right)^2 + \frac{-1 \cdot 1 \cdot 3}{3!} \left(\frac{1}{4}\right)^3 + \dots \\ &= 1 + \frac{p}{1!} \left(\frac{x}{q}\right) + \frac{p(p+q)}{2!} \left(\frac{x}{q}\right)^2 + \frac{p(p+q)(p+2q)}{3!} \left(\frac{x}{q}\right)^3 + \dots, \end{aligned}$$

where $\boxed{p = -1}$; $p + q = 1 \Rightarrow \boxed{q = 2}$ and $\frac{x}{q} = \frac{1}{4} \Rightarrow \boxed{x = \frac{2}{4} = \frac{1}{2}}$.

$$\Rightarrow 1 - S = (1 - x)^{-p/q} = \left(1 - \frac{1}{2}\right)^{-(-1)/2} = \left(\frac{1}{2}\right)^{1/2} \Rightarrow S = 1 - \left(\frac{1}{2}\right)^{1/2} \Rightarrow \boxed{S = 1 - \frac{1}{\sqrt{2}}}$$

Problem 10.6

Sum to ∞ the series $\frac{3}{18} + \frac{3 \cdot 7}{18 \cdot 24} + \frac{3 \cdot 7 \cdot 11}{18 \cdot 24 \cdot 30} + \dots$

Solution. Let

$$S = \frac{3}{18} + \frac{3 \cdot 7}{18 \cdot 24} + \frac{3 \cdot 7 \cdot 11}{18 \cdot 24 \cdot 30} + \dots = \frac{3}{3 \cdot 6} + \frac{3 \cdot 7}{(3 \cdot 6)(4 \cdot 6)} + \frac{3 \cdot 7 \cdot 11}{(3 \cdot 6)(4 \cdot 6)(5 \cdot 6)} + \dots$$

Dividing by $(1 \cdot 6)(2 \cdot 6) = 72$:

$$\frac{S}{(1 \cdot 6)(2 \cdot 6)} = \frac{3}{(1 \cdot 6)(2 \cdot 6)(3 \cdot 6)} + \frac{3 \cdot 7}{(1 \cdot 6)(2 \cdot 6)(3 \cdot 6)(4 \cdot 6)} + \dots \Rightarrow \frac{S}{72} = \frac{3}{3!6^3} + \frac{3 \cdot 7}{4!6^4} + \frac{3 \cdot 7 \cdot 11}{5!6^5} + \dots$$

To match the factors $p(p+q)(p+2q) \dots$ with $q = 4$, multiply by $(-5)(-1)$:

$$\frac{(-5)(-1)S}{72} = \frac{(-5)(-1)(3)}{3!} \left(\frac{1}{6}\right)^3 + \frac{(-5)(-1)(3)(7)}{4!} \left(\frac{1}{6}\right)^4 + \frac{(-5)(-1)(3)(7)(11)}{5!} \left(\frac{1}{6}\right)^5 + \dots$$

Adding the missing first terms $1 + \frac{-5}{1!} \left(\frac{1}{6}\right) + \frac{(-5)(-1)}{2!} \left(\frac{1}{6}\right)^2$ on both sides:

$$1 + \frac{-5}{1!} \left(\frac{1}{6}\right) + \frac{(-5)(-1)}{2!} \left(\frac{1}{6}\right)^2 + \frac{5S}{72} = 1 + \frac{p}{1!} \left(\frac{x}{q}\right) + \frac{p(p+q)}{2!} \left(\frac{x}{q}\right)^2 + \frac{p(p+q)(p+2q)}{3!} \left(\frac{x}{q}\right)^3 + \dots,$$

where $\boxed{p = -5}$, $p + q = -1 \Rightarrow \boxed{q = 4}$, $\frac{x}{q} = \frac{1}{6} \Rightarrow \boxed{x = \frac{4}{6} = \frac{2}{3}}$. Now LHS $= 1 - \frac{5}{6} + \frac{5}{72} + \frac{5S}{72} = \frac{72-60+5}{72} + \frac{5S}{72} = \frac{17}{72} + \frac{5S}{72}$,

and RHS $= (1 - x)^{-p/q} = \left(1 - \frac{2}{3}\right)^{5/4} = \left(\frac{1}{3}\right)^{5/4}$.

$$\Rightarrow \frac{5S}{72} = \left(\frac{1}{3}\right)^{5/4} - \frac{17}{72} \Rightarrow \boxed{S = \frac{72}{5} \left(\left(\frac{1}{3}\right)^{5/4} - \frac{17}{72} \right)}$$

Problem 10.7 (Exercise)Sum the following series to ∞ :

$$(i) \frac{4}{20} + \frac{4 \cdot 7}{20 \cdot 30} + \frac{4 \cdot 7 \cdot 10}{20 \cdot 30 \cdot 40} + \cdots \quad (ii) \frac{1}{24} - \frac{1 \cdot 3}{24 \cdot 32} + \frac{1 \cdot 3 \cdot 5}{24 \cdot 32 \cdot 40} - \cdots = \frac{1}{3} (179 - 80\sqrt{5}).$$

Solution. (i) Writing $20 = 2 \cdot 10$, $30 = 3 \cdot 10$, $40 = 4 \cdot 10$ and dividing by $1 \cdot 10$:

$$\begin{aligned} \frac{S}{10} &= \frac{4}{2!10^2} + \frac{4 \cdot 7}{3!10^3} + \frac{4 \cdot 7 \cdot 10}{4!10^4} + \cdots \\ \Rightarrow \frac{1 \cdot S}{10} &= \frac{1 \cdot 4}{2!} \left(\frac{1}{10}\right)^2 + \frac{1 \cdot 4 \cdot 7}{3!} \left(\frac{1}{10}\right)^3 + \frac{1 \cdot 4 \cdot 7 \cdot 10}{4!} \left(\frac{1}{10}\right)^4 + \cdots \\ \Rightarrow 1 + \frac{1}{1!} \left(\frac{1}{10}\right) + \frac{S}{10} &= 1 + \frac{1}{1!} \left(\frac{1}{10}\right) \\ &\quad + \frac{1 \cdot 4}{2!} \left(\frac{1}{10}\right)^2 + \frac{1 \cdot 4 \cdot 7}{3!} \left(\frac{1}{10}\right)^3 + \cdots \end{aligned}$$

with $p = 1$, $q = 3$, $x = \frac{3}{10}$:

$$\frac{11}{10} + \frac{S}{10} = \left(1 - \frac{3}{10}\right)^{-1/3} = \left(\frac{7}{10}\right)^{-1/3} = \left(\frac{10}{7}\right)^{1/3} \Rightarrow \boxed{S = 10 \left(\frac{10}{7}\right)^{1/3} - 11 = 10 \sqrt[3]{\frac{10}{7}} - 11}$$

(ii) Proceed similarly (here $24 = 3 \cdot 8$, $32 = 4 \cdot 8$, $40 = 5 \cdot 8$, and $(1+x)^{-p/q}$ is used since the signs alternate). ■

11 Exponential series

Result 11.1 (Exponential series)

$$\begin{aligned} e^x &= 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} + \cdots \\ e^{-x} &= 1 - \frac{x}{1!} + \frac{x^2}{2!} - \frac{x^3}{3!} + \cdots + (-1)^n \frac{x^n}{n!} + \cdots \\ \frac{e^x + e^{-x}}{2} &= 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots, \quad \frac{e^x - e^{-x}}{2} = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots \end{aligned}$$

In particular,

$$\begin{aligned} e &= 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots, & e^{-1} &= 1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \cdots, \\ \frac{e + e^{-1}}{2} &= 1 + \frac{1}{2!} + \frac{1}{4!} + \cdots, & \frac{e - e^{-1}}{2} &= 1 + \frac{1}{3!} + \frac{1}{5!} + \cdots \end{aligned}$$

Note**Remark.** Suppose we have the function a^x . Let $y = a^x \Rightarrow \log_e y = \log_e a^x = x \log_e a \Rightarrow y = e^{x \log_e a}$

$$\Rightarrow a^x = e^{(\log_e a)x} \Rightarrow a^x = 1 + \frac{(\log_e a)x}{1!} + \frac{(\log_e a)^2 x^2}{2!} + \frac{(\log_e a)^3 x^3}{3!} + \cdots \quad \left(\because e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \cdots\right).$$

11.1 Problems

Problem 11.1Find the coefficient of x^n in $(1+2x)e^{-2x}$.

Solution.

$$\begin{aligned}(1+2x)e^{-2x} &= (1+2x) \left[1 - \frac{2x}{1!} + \frac{(2x)^2}{2!} - \frac{(2x)^3}{3!} + \cdots + \frac{(-1)^n(2x)^n}{n!} + \cdots \right] \\ &= (1+2x) \left(1 - \frac{2}{1!}x + \frac{2^2}{2!}x^2 - \frac{2^3}{3!}x^3 + \cdots + \frac{(-1)^{n-1}2^{n-1}}{(n-1)!}x^{n-1} + \frac{(-1)^n2^n}{n!}x^n + \cdots \right)\end{aligned}$$

$$\begin{aligned}\Rightarrow \text{Coefficient of } x^n \text{ in } (1+2x)e^{-2x} &= \frac{(-1)^n2^n}{n!} + 2 \cdot \frac{(-1)^{n-1}2^{n-1}}{(n-1)!} = \frac{(-1)^n2^n}{n!} + \frac{(-1)^{n-1}2^n}{(n-1)!} \\ &= \frac{(-1)^n2^n}{n!} [1 + (-1)^{-1}n] \Rightarrow \boxed{\text{Coeff. of } x^n = \frac{(-1)^n2^n}{n!}(1-n)}\end{aligned}$$

■

Problem 11.2

Sum the series to ∞ : $\frac{1}{2!} + \frac{1+2}{3!} + \frac{1+2+2^2}{4!} + \cdots$

Solution. Let $S = \frac{1}{2!} + \frac{1+2}{3!} + \frac{1+2+2^2}{4!} + \cdots = T_1 + T_2 + T_3 + \cdots = \sum_{n=1}^{\infty} T_n$, where

$$\begin{aligned}T_n &= \frac{1+2+2^2+\cdots+2^{n-1}}{(n+1)!} = \frac{(2^n-1)/(2-1)}{(n+1)!} \quad \left(\because 1+r+r^2+\cdots+r^{n-1} = \frac{r^n-1}{r-1} \text{ if } r > 1 \right) \\ \Rightarrow T_n &= \frac{2^n-1}{(n+1)!} = \frac{2^n}{(n+1)!} - \frac{1}{(n+1)!}.\end{aligned}$$

$$\begin{aligned}\therefore S &= \sum_{n=1}^{\infty} T_n = \sum_{n=1}^{\infty} \frac{2^n}{(n+1)!} - \sum_{n=1}^{\infty} \frac{1}{(n+1)!} \\ &= \left(\frac{2}{2!} + \frac{2^2}{3!} + \frac{2^3}{4!} + \cdots \right) - \left(\frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \cdots \right) \\ &= \frac{1}{2} \left[\frac{2^2}{2!} + \frac{2^3}{3!} + \frac{2^4}{4!} + \cdots \right] - \left[\left(1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots \right) - 1 - \frac{1}{1!} \right] \\ &= \frac{1}{2} \left[\left(1 + \frac{2}{1!} + \frac{2^2}{2!} + \frac{2^3}{3!} + \cdots \right) - 1 - \frac{2}{1!} \right] - (e-2) \\ &= \frac{1}{2} (e^2-3) - e + 2 = \frac{1}{2}e^2 - e + \frac{1}{2} = \frac{e^2-2e+1}{2} \Rightarrow \boxed{S = \frac{(e-1)^2}{2}}\end{aligned}$$

■

Problem 11.3

Sum the series to ∞ : $\frac{1}{1!} + \frac{3}{2!} + \frac{5}{3!} + \frac{7}{4!} + \cdots$

Solution. Let $S = \frac{1}{1!} + \frac{3}{2!} + \frac{5}{3!} + \frac{7}{4!} + \cdots = T_1 + T_2 + T_3 + \cdots = \sum_{n=1}^{\infty} T_n$, where

$$T_n = \frac{2n-1}{n!} = \frac{2n}{n!} - \frac{1}{n!} = \frac{2}{(n-1)!} - \frac{1}{n!}.$$

$$\begin{aligned}\therefore S &= \sum_{n=1}^{\infty} T_n = \sum_{n=1}^{\infty} \left(\frac{2}{(n-1)!} - \frac{1}{n!} \right) = 2 \sum_{n=1}^{\infty} \frac{1}{(n-1)!} - \sum_{n=1}^{\infty} \frac{1}{n!} \\ &= 2 \left[1 + \frac{1}{1!} + \frac{1}{2!} + \cdots \right] - \left(\frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots \right) = 2e - (e-1) \Rightarrow \boxed{S = e+1}\end{aligned}$$

■

Problem 11.4

Sum the series $\frac{3^2}{1!} + \frac{5^2}{2!} + \frac{7^2}{3!} + \dots$

Solution. Let $S = \frac{3^2}{1!} + \frac{5^2}{2!} + \frac{7^2}{3!} + \dots = \sum_{n=1}^{\infty} T_n$, where

$$T_n = \frac{(2n+1)^2}{n!} = \frac{4n^2 + 4n + 1}{n!}.$$

Now write $4n^2 + 4n + 1 = A + Bn + Cn(n-1)$ (since $n! = n(n-1)(n-2)\dots 1$).

Put $n = 0$: $\boxed{1 = A}$. Put $n = 1$: $9 = A + B \Rightarrow \boxed{B = 8}$. Compare n^2 : $\boxed{4 = C}$.

$$\therefore 4n^2 + 4n + 1 = 1 + 8n + 4n(n-1) \Rightarrow T_n = \frac{1 + 8n + 4n(n-1)}{n!} = \frac{1}{n!} + \frac{8}{(n-1)!} + \frac{4}{(n-2)!}.$$

$$\begin{aligned} \therefore S &= \sum_{n=1}^{\infty} T_n = \sum_{n=1}^{\infty} \frac{1}{n!} + 8 \sum_{n=1}^{\infty} \frac{1}{(n-1)!} + 4 \sum_{n=1}^{\infty} \frac{1}{(n-2)!} \\ &= \left(\frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots \right) + 8 \left(1 + \frac{1}{1!} + \frac{1}{2!} + \dots \right) + 4 \left(1 + \frac{1}{1!} + \frac{1}{2!} + \dots \right) \end{aligned}$$

(for $n = 1$, the term $\frac{1}{(-1)!}$ is neglected — negative factorials are taken as zero)

$$= (e - 1) + 8e + 4e \Rightarrow \boxed{S = 13e - 1}$$

■

Problem 11.5

Sum the series to ∞ : $\frac{1 \cdot 2 \cdot 3}{2!} + \frac{2 \cdot 4 \cdot 5}{3!} + \frac{3 \cdot 6 \cdot 7}{4!} + \dots$

Solution. Let $S = \frac{1 \cdot 2 \cdot 3}{2!} + \frac{2 \cdot 4 \cdot 5}{3!} + \frac{3 \cdot 6 \cdot 7}{4!} + \dots = \sum_{n=1}^{\infty} T_n$, where

$$T_n = \frac{n(2n)(2n+1)}{(n+1)!} = \frac{2n^2(2n+1)}{(n+1)!} \Rightarrow T_n = \frac{4n^3 + 2n^2}{(n+1)!}.$$

Now write $4n^3 + 2n^2 = A + B(n+1) + C(n+1)n + D(n+1)n(n-1)$ (since $(n+1)! = (n+1)n(n-1)\dots$).

Put $n = -1$: $\boxed{-2 = A}$. Put $n = 0$: $0 = A + B \Rightarrow \boxed{B = 2}$. Put $n = 1$: $6 = A + 2B + 2C \Rightarrow C = \frac{1}{2}(6 - A - 2B) \Rightarrow \boxed{C = 2}$. Compare n^3 : $\boxed{4 = D}$.

$$\Rightarrow 4n^3 + 2n^2 = -2 + 2(n+1) + 2(n+1)n + 4(n+1)n(n-1)$$

$$\begin{aligned} \therefore S &= \sum_{n=1}^{\infty} T_n = \sum_{n=1}^{\infty} \frac{-2 + 2(n+1) + 2(n+1)n + 4(n+1)n(n-1)}{(n+1)!} \\ &\Rightarrow S = -2 \sum_{n=1}^{\infty} \frac{1}{(n+1)!} + 2 \sum_{n=1}^{\infty} \frac{1}{n!} + 2 \sum_{n=1}^{\infty} \frac{1}{(n-1)!} + 4 \sum_{n=1}^{\infty} \frac{1}{(n-2)!} \\ &= -2 \left(\frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \dots \right) + 2 \left(\frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots \right) \\ &\quad + 2 \left(1 + \frac{1}{1!} + \frac{1}{2!} + \dots \right) + 4 \left(1 + \frac{1}{1!} + \frac{1}{2!} + \dots \right) \\ &= -2(e - 2) + 2(e - 1) + 2e + 4e = -2e + 4 + 2e - 2 + 2e + 4e \Rightarrow \boxed{S = 6e + 2} \end{aligned}$$

■

Exercise. Sum the series to ∞ : $\frac{1^3}{1!} + \frac{2^3}{2!} + \frac{3^3}{3!} + \dots$ (**Ans:** $5e$; write $n^3 = n + 3n(n-1) + n(n-1)(n-2)$.)

12 Logarithmic series

Result 12.1 (Logarithmic series, $|x| < 1$)

$$\log_e(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots, \quad \log_e(1-x) = -\left(x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \dots\right).$$

Then

$$\begin{aligned} \log_e(1+x) + \log_e(1-x) &= -2\left(\frac{x^2}{2} + \frac{x^4}{4} + \frac{x^6}{6} + \dots\right) \Rightarrow \log_e(1+x)(1-x) = \log(1-x^2) = -2\left(\frac{x^2}{2} + \frac{x^4}{4} + \dots\right) \\ &\Rightarrow \boxed{\frac{x^2}{2} + \frac{x^4}{4} + \frac{x^6}{6} + \dots = -\frac{1}{2}\log(1-x^2)} \end{aligned}$$

Similarly

$$\begin{aligned} \log_e(1+x) - \log_e(1-x) &= 2\left(x + \frac{x^3}{3} + \frac{x^5}{5} + \dots\right) \Rightarrow \log_e\left(\frac{1+x}{1-x}\right) = 2\left(x + \frac{x^3}{3} + \frac{x^5}{5} + \dots\right) \\ &\Rightarrow \boxed{x + \frac{x^3}{3} + \frac{x^5}{5} + \dots = \frac{1}{2}\log_e\left(\frac{1+x}{1-x}\right)} \quad \left(\text{also we know } \tanh^{-1} x = \frac{1}{2}\log_e\left(\frac{1+x}{1-x}\right)\right). \end{aligned}$$

Note

Remark. If we put $x = 1$ in $\log_e(1+x)$, we get

$$\log_e 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \dots$$

12.1 Problems

Problem 12.1

Show that $\frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{3 \cdot 4 \cdot 5} + \frac{1}{5 \cdot 6 \cdot 7} + \dots = \log_e 2 - \frac{1}{2}$.

Solution. Let $S = \frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{3 \cdot 4 \cdot 5} + \frac{1}{5 \cdot 6 \cdot 7} + \dots = T_1 + T_2 + T_3 + \dots = \sum_{n=1}^{\infty} T_n$, where

$$T_n = \frac{1}{(2n-1)(2n)(2n+1)}.$$

Now (partial fractions)

$$\begin{aligned} \frac{1}{(2n-1)(2n)(2n+1)} &= \frac{A}{2n-1} + \frac{B}{2n} + \frac{C}{2n+1} = \frac{A(2n)(2n+1) + B(2n-1)(2n+1) + C(2n-1)(2n)}{(2n-1)(2n)(2n+1)} \\ &\Rightarrow 1 = A(2n)(2n+1) + B(2n-1)(2n+1) + C(2n-1)(2n). \end{aligned}$$

$$\begin{aligned} \text{Put } n = \frac{1}{2}: 1 = A(1)(2) &\Rightarrow \boxed{A = \frac{1}{2}}. \quad \text{Put } n = 0: 1 = B(-1)(1) \Rightarrow \boxed{B = -1}. \quad \text{Put } n = -\frac{1}{2}: 1 = C(-2)(-1) \Rightarrow \\ &\boxed{C = \frac{1}{2}}. \end{aligned}$$

$$\text{Then } T_n = \frac{A}{2n-1} + \frac{B}{2n} + \frac{C}{2n+1} = \frac{1}{2}\left(\frac{1}{2n-1}\right) - \frac{1}{2n} + \frac{1}{2}\left(\frac{1}{2n+1}\right).$$

$$\begin{aligned}
\text{But } S &= \sum_{n=1}^{\infty} T_n = \frac{1}{2} \sum_{n=1}^{\infty} \frac{1}{2n-1} - \sum_{n=1}^{\infty} \frac{1}{2n} + \frac{1}{2} \sum_{n=1}^{\infty} \frac{1}{2n+1} \\
&= \frac{1}{2} \left(\frac{1}{1} + \frac{1}{3} + \frac{1}{5} + \dots \right) - \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \dots \right) + \frac{1}{2} \left(\frac{1}{3} + \frac{1}{5} + \frac{1}{7} + \dots \right) \\
&= \frac{1}{2} - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \dots \\
&= \frac{1}{2} + \left(-\frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \dots \right) \\
&= \frac{1}{2} + (\log_e 2 - 1) \Rightarrow \boxed{S = \log_e 2 - \frac{1}{2}}
\end{aligned}$$

■

Problem 12.2

Sum the series to infinity: $\frac{5}{1 \cdot 2 \cdot 3} + \frac{7}{3 \cdot 4 \cdot 5} + \frac{9}{5 \cdot 6 \cdot 7} + \dots$

Solution. Let $S = \frac{5}{1 \cdot 2 \cdot 3} + \frac{7}{3 \cdot 4 \cdot 5} + \frac{9}{5 \cdot 6 \cdot 7} + \dots = T_1 + T_2 + T_3 + \dots = \sum_{n=1}^{\infty} T_n$, where

$$T_n = \frac{2n+3}{(2n-1)(2n)(2n+1)} = \frac{A}{2n-1} + \frac{B}{2n} + \frac{C}{2n+1}.$$

Now $2n+3 = A(2n)(2n+1) + B(2n-1)(2n+1) + C(2n-1)(2n)$.

Put $n = \frac{1}{2}$: $4 = A(1)(2) \Rightarrow \boxed{A = 2}$. Put $n = 0$: $3 = B(-1)(1) \Rightarrow \boxed{B = -3}$. Put $n = -\frac{1}{2}$: $2 = C(-2)(-1) \Rightarrow \boxed{C = 1}$.

$$\Rightarrow T_n = 2 \left(\frac{1}{2n-1} \right) - 3 \left(\frac{1}{2n} \right) + \frac{1}{2n+1}.$$

$$\begin{aligned}
\text{But } S &= \sum_{n=1}^{\infty} T_n = 2 \sum_{n=1}^{\infty} \frac{1}{2n-1} - 3 \sum_{n=1}^{\infty} \frac{1}{2n} + \sum_{n=1}^{\infty} \frac{1}{2n+1} \\
&= 2 \left(\frac{1}{1} + \frac{1}{3} + \frac{1}{5} + \dots \right) - 3 \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{6} + \dots \right) + \left(\frac{1}{3} + \frac{1}{5} + \frac{1}{7} + \dots \right) \\
&= 2 - 3 \left(\frac{1}{2} \right) + 3 \left(\frac{1}{3} \right) - 3 \left(\frac{1}{4} \right) + 3 \left(\frac{1}{5} \right) - \dots \\
&= 2 + 3 \left(-\frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \dots \right) \\
&= 2 + 3 (\log_e 2 - 1) \Rightarrow \boxed{S = 3 \log_e 2 - 1 = \log_e 8 - 1}
\end{aligned}$$

■

Problem 12.3

Prove that $\log_e \sqrt{12} = 1 + \left(\frac{1}{2} + \frac{1}{3} \right) \frac{1}{4} + \left(\frac{1}{4} + \frac{1}{5} \right) \frac{1}{4^2} + \dots$

Solution.

$$\begin{aligned}
\text{RHS} &= 1 + \left(\frac{1}{2} + \frac{1}{3} \right) \frac{1}{4} + \left(\frac{1}{4} + \frac{1}{5} \right) \frac{1}{4^2} + \dots \\
&= 1 + \frac{1}{2} \left(\frac{1}{4} \right) + \frac{1}{3} \left(\frac{1}{4} \right) + \frac{1}{4} \left(\frac{1}{4} \right)^2 + \frac{1}{5} \left(\frac{1}{4} \right)^2 + \dots \\
&= 1 + \frac{1}{2} \left(\frac{1}{2} \right)^2 + \frac{1}{3} \left(\frac{1}{2} \right)^2 + \frac{1}{4} \left(\left(\frac{1}{2} \right)^2 \right)^2 + \frac{1}{5} \left(\left(\frac{1}{2} \right)^2 \right)^2 + \dots \\
&= \left[\frac{1}{2} \left(\frac{1}{2} \right)^2 + \frac{1}{4} \left(\frac{1}{2} \right)^4 + \dots \right] + \left[1 + \frac{1}{3} \left(\frac{1}{2} \right)^2 + \frac{1}{5} \left(\frac{1}{2} \right)^4 + \dots \right] \\
&= \left[\frac{1}{2} \left(\frac{1}{2} \right)^2 + \frac{1}{4} \left(\frac{1}{2} \right)^4 + \dots \right] + 2 \left[\frac{1}{2} + \frac{1}{3} \left(\frac{1}{2} \right)^3 + \frac{1}{5} \left(\frac{1}{2} \right)^5 + \dots \right]
\end{aligned}$$

$$\begin{aligned}
&= \left(\frac{x^2}{2} + \frac{x^4}{4} + \cdots \right) + 2 \left(x + \frac{x^3}{3} + \frac{x^5}{5} + \cdots \right), \quad \text{when } x = \frac{1}{2} \\
&= -\frac{1}{2} \log(1 - x^2) + \log_e \left(\frac{1+x}{1-x} \right) \\
&= -\frac{1}{2} \log_e \left(1 - \frac{1}{4} \right) + \log_e \left(\frac{3/2}{1/2} \right) \\
&= -\frac{1}{2} \log_e \left(\frac{3}{4} \right) + \log_e 3 = \log_e \left(\frac{3}{4} \right)^{-1/2} + \log_e 3 = \log_e \left(\frac{\sqrt{4}}{\sqrt{3}} \right) + \log_e 3 \\
&= \log_e \left(\frac{\sqrt{4}}{\sqrt{3}} \times 3 \right) = \log_e \left(\frac{\sqrt{4}}{\sqrt{3}} \times \sqrt{3}\sqrt{3} \right) = \log_e \sqrt{12} = \text{LHS}.
\end{aligned}$$

■

Problem 12.4 (Exercise)

Sum to infinity the series $\left(1 + \frac{1}{2}\right) + \left(\frac{1}{3} + \frac{1}{4}\right) \left(\frac{1}{9}\right) + \left(\frac{1}{5} + \frac{1}{6}\right) \frac{1}{9^2} + \cdots$

Solution.

$$\begin{aligned}
S &= 1 + \frac{1}{2} + \frac{1}{3} \left(\frac{1}{3}\right)^2 + \frac{1}{4} \left(\frac{1}{3}\right)^2 + \frac{1}{5} \left(\frac{1}{3}\right)^4 + \frac{1}{6} \left(\frac{1}{3}\right)^4 + \cdots \\
&= \left(\frac{1}{2} + \frac{1}{4} \left(\frac{1}{3}\right)^2 + \frac{1}{6} \left(\frac{1}{3}\right)^4 + \cdots \right) + \left(1 + \frac{1}{3} \left(\frac{1}{3}\right)^2 + \frac{1}{5} \left(\frac{1}{3}\right)^4 + \cdots \right) \\
&= 3^2 \left(\frac{1}{2} \left(\frac{1}{3}\right)^2 + \frac{1}{4} \left(\frac{1}{3}\right)^4 + \frac{1}{6} \left(\frac{1}{3}\right)^6 + \cdots \right) + 3 \left(\frac{1}{3} + \frac{1}{3} \left(\frac{1}{3}\right)^3 + \frac{1}{5} \left(\frac{1}{3}\right)^5 + \cdots \right) \\
&= 9 \left(\frac{x^2}{2} + \frac{x^4}{4} + \cdots \right) + 3 \left(x + \frac{x^3}{3} + \frac{x^5}{5} + \cdots \right), \quad x = \frac{1}{3} \\
&= -\frac{9}{2} \log_e \left(1 - \frac{1}{9} \right) + \frac{3}{2} \log_e \left(\frac{4/3}{2/3} \right) \Rightarrow \boxed{S = \frac{9}{2} \log_e \frac{9}{8} + \frac{3}{2} \log_e 2}
\end{aligned}$$

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Assignment – III (Theory of Equations & Algebra)

- (1) Solve the equation $x^5 - 3x^4 + 4x^3 + 4x^2 - 21x + 15 = 0$, given that $1 + 2i$ and $\sqrt{3}$ are two roots.
- (2) Solve the following equations:
 - (i) $x^4 + 2x^3 - 21x^2 - 22x + 40 = 0$, whose roots are in A.P.
 - (ii) $x^4 - 15x^3 + 70x^2 - 120x + 64 = 0$, given that the roots are in Geometrical Progression.
 - (iii) $40x^4 - 22x^3 - 21x^2 + 2x + 1 = 0$, given that the roots are in Harmonical Progression.
- (3) If the equation $x^4 - 4x^3 - 2x^2 + 12x + 9 = 0$ has two pairs of equal roots, find them.
- (4) Solve $x^4 + 4x^3 - 5x^2 - 8x + 6 = 0$, given that the sum of two of its roots is zero.
- (5) Form the fourth degree equation two of whose roots are $1 + i$ and $2 + \sqrt{3}$.
- (6) Solve $3x^3 - 32x^2 + 33x + 108 = 0$, given that one root is the square of another.
- (7) Solve $x^4 - 4x^3 - 17x^2 + 24x + 36 = 0$, given that the product of two of the roots is 12.
- (8) If α, β, γ are the roots of $x^3 - 3ax + b = 0$, show that $\sum(\alpha - \beta)(\alpha - \gamma) = 9a$.
- (9) If α, β, γ are the roots of $x^3 + 2x^2 + 3x + 3 = 0$, prove that $\sum \frac{\alpha^2}{(\alpha + 1)^2} = 13$.
- (10) If α, β, γ are the roots of $x^3 + px + q = 0$, form the equation whose roots are $(\alpha - \beta)^2, (\beta - \gamma)^2, (\gamma - \alpha)^2$.

- (11) Diminish the roots of $x^4 - 8x^3 + 19x^2 - 12x + 2 = 0$ by 2 and hence solve the equation.
- (12) Increase by 7 the roots of the equation $3x^4 + 7x^3 - 15x^2 + x - 2 = 0$.
- (13) Remove the second term in $x^4 - 12x^3 + 48x^2 - 72x + 35 = 0$ and hence solve it.
- (14) Solve $8x^4 + 4x^3 - 18x^2 + 11x - 2 = 0$, given that it has three equal roots. [**Hint:** Solve $f''(x) = 0$ and proceed.]
- (15) Find k , if the equation $2x^3 - 9x^2 + 12x + k = 0$ has a double root.
- (16) Solve the following equations: (i) $3x^6 + x^5 - 27x^4 + 27x^2 - x - 3 = 0$ (ii) $x^5 - x^4 + x^3 - x^2 + x - 1 = 0$.
- (17) Determine the stability of the closed loop system with forward transfer function

$$G(s) = \frac{128}{s(s^7 + 3s^6 + 10s^5 + 24s^4 + 48s^3 + 96s^2 + 128s + 192)}$$

and unity feedback, from its characteristic equation

$$s^8 + 3s^7 + 10s^6 + 24s^5 + 48s^4 + 96s^3 + 128s^2 + 192s + 128 = 0.$$

- (18) Sum the following series to ∞ :

$$\begin{aligned} \text{(i)} \quad & \frac{5}{1!} + \frac{7}{3!} + \frac{9}{5!} + \dots & \text{(ii)} \quad & \frac{1^3x}{1!} + \frac{2^3x^2}{2!} + \frac{3^3x^3}{3!} + \dots & \text{(iii)} \quad & \frac{1}{1 \cdot 2 \cdot 3} + \frac{5}{3 \cdot 4 \cdot 5} + \frac{9}{5 \cdot 6 \cdot 7} + \dots \\ \text{(iv)} \quad & \frac{1}{1 \cdot 2 \cdot 3} + \frac{1}{5 \cdot 6 \cdot 7} + \frac{1}{9 \cdot 10 \cdot 11} + \dots & \text{(v)} \quad & \frac{5}{12} + \frac{5 \cdot 7}{12 \cdot 16} + \frac{5 \cdot 7 \cdot 9}{12 \cdot 16 \cdot 20} + \dots \end{aligned}$$